Coupling a multi-species point model to a continuum biofilm growth model: verification and the effect of the composition on growth and stress

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Abstract

Continuum models of biofilm growth predict where biomass forms and the stresses it causes, but treat the biofilm as one material; multi-species point models describe composition, living fraction and antibiotic response, but have no space. This work couples a published variational multi-species point model to a Hamilton-principle growth model at every integration point without changing either model. The point model receives the local nutrient and antibiotic, and its living volume fraction sets their consumption. In this coupling the composition reaches growth and stress only through the nutrient and the front speed of the growth equation, because the growth law counts all biomass: the growth is bounded by the growth rate times the time, and the front speed changes by at most a factor of two. Two-species examples, computed in a finite-volume code and in user elements of ANSYS and Abaqus and verified by a manufactured solution and grid convergence indices up to 192^3 cells, show this: an antibiotic lowers the living fraction of the seed from 0.96 to 0.15, or to 0.04 when the species start mixed, while the growth changes by 2 % or less and the converged seed stress by no more than its grid uncertainty (2 % when the species start apart, 7 % against a band of 9 % when they start mixed). The nutrient decides where the antibiotic acts, and the state given to newly reached material decides in a maze whether the biofilm recovers. A growth law derived from the energy that counts only living biomass lowers the seed growth under the antibiotic by 49 to 66 % and its stress by 16 to 30 %.

Keywords: biofilm growth; multi-species model; antibiotic; extended Hamilton principle; code verification; growth-induced stress

1 Introduction

Oral biofilms on teeth and implants are communities of several bacterial species. Their composition changes over days: early colonisers prepare the surface for later ones, and pathogenic species appear only when the community allows it (Heine et al., 2025). Treatments such as antibiotics act on the composition, while the mechanical load on the surface comes from the amount and the growth of the biomass. A model that predicts the stresses for a given treatment therefore has to describe both: where the biofilm grows, and what it is made of.

Two kinds of model exist for this. Continuum growth models describe the amount of biomass in space. Wanner and Gujer (1986) introduced the one-dimensional multi-species biofilm model, Eberl et al. (2001) a density-dependent diffusion model of the spreading, and Alpkvist and Klapper (2007) and Rahman et al. (2015) multi-dimensional multi-species continuum models in which each species is transported. Feng et al. (2021) model a two-species oral biofilm in which the spreading follows the summed growth of the species, and Soleimani et al. (2023) the

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co-aggregation of two species in a user element. Klempt et al. (2024) derive the growth of a biofilm from the extended Hamilton principle (Junker and Balzani, 2021), with a phase field ϕ_{3D}, a nutrient c and the growth variable α of a multiplicative split $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$ (the multiplicative split of growth goes back to Rodriguez et al., 1994). From it they compute the resulting stresses; their model has one biomass and no composition. Growth-induced stresses also shape bacterial colonies in experiments (Seminara et al., 2012). Point models describe the composition of a community at one point. Klempt et al. (2026) derive a multi-species model from the same variational principle, with volume fractions ϕ_i and living fractions ψ_i of N species, an interaction matrix $\mathbf{A}$ and the response of each species to an antibiotic. Fritsch et al. (2025) calibrate its parameters by Bayesian updating. These models have no space: the nutrient c^* and the antibiotic α^* are given inputs.

The point model needs local values of c^* and α^*, which a growth field can supply. The growth field needs to know how much of the biomass is alive, because only living biomass consumes nutrient. Connecting the two at every material point is the natural step, but it raises questions that neither paper answers. These questions are which variables are passed, whether either model has to be changed, which new assumptions are needed (for example the composition of biofilm that a point reaches later), and whether the composition reaches the growth and the stress at all.

This work couples the point model of Klempt et al. (2026) to the growth model of Klempt et al. (2024). Unlike the multi-species continuum models above, the composition is computed by a variational point model used as published, without transport of the species, and the coupling is made without changing either set of equations. The point model runs at every integration point as published, receives $c^* = c_0^* c$ and $\alpha^* = \alpha_0^* a$ from the nutrient and antibiotic fields and returns the living volume fraction $\sum_i \bar{\phi}_i$. This sets the consumption $g\,\phi_{3D}(\sum_i \bar{\phi}_i)\,c$ in the field equations. The contributions are the following.

1. With this minimal, equation-preserving coupling the composition is nearly silent in the mechanics, and the reason can be stated. In the model of Klempt et al. (2024) the composition reaches the growth and the stress along one path only, through the nutrient and the front speed of the growth equation, because the growth law counts all biomass. The growth is bounded by $k_\alpha T^*$ and the front speed changes by at most a factor of two (Section 2.4). The examples quantify it: the antibiotic removes most of the living biomass of the seed, while the growth changes by 2 % or less and the converged seed stress by no more than its grid uncertainty. Any model in which the composition is to change the mechanics therefore needs a growth rate or a front speed tied to the living biomass or to the species. The coupling presented here is the base on which such a term can be added, and the size of its effect is estimated from the runs (Section 5).

2. Three further properties are derived from the published equations and read against the examples. The first is the fast adjustment of the nutrient, with its Thiele modulus. The second is the balance by which the antibiotic acts in the point model, whose decay rate grows as the inverse square of the volume fraction of a species. The third is the time link, which is equivalent to a scaling of the viscosities and is therefore fixed by the data.

3. The coupling is implemented independently in a finite-volume code and in user elements of ANSYS and Abaqus. All three reproduce the point model alone; the finite-volume and the Galerkin code are verified by a manufactured solution and grid convergence indices, and the neighboured element of ANSYS agrees with them within its larger grid-convergence band. The front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ of the growth equation sets the accuracy of the stress and needs a second-order treatment.

4. The state that newly reached material receives is fixed by neither model. It is stated as an open modelling choice, and the examples show its effect: in a maze it decides whether the biofilm recovers after the antibiotic.

The examples use two species and an antibiotic, with the synthetic data of Fritsch et al. (2025) and the constants of Klempt et al. (2024, 2026), on a cube, against obstacles, in a maze and around the crown of a human molar. Section 2 states the models, the coupling and what its structure implies, Section 3 the implementations and their verification in short (details in the appendix), Section 4 the examples, Section 5 the limits of the coupling, and Section 6 concludes.

2 Models and coupling

2.1 Growth field

The biofilm is described by the phase field $\phi_{3D} \in [0,1]$, the nutrient c and the growth variable α of Klempt et al. (2024), with $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$, $\mathbf{F}_g = \alpha\mathbf{I}$ and $\alpha(0) = 1$, so that the growth is $\alpha - 1$. After division by the viscosities, their Eqs. 34 to 36 read

$$\dot{\phi_{3D}} = \beta\,\nabla^2\phi_{3D} + k_\alpha\,\alpha - r\,\frac{c}{k+c}\,\mathbf{n}_c\cdot\nabla\phi_{3D}, \qquad \mathbf{n}_c = \frac{\nabla c}{|\nabla c|}, \tag{1}$$

$$\dot{c} = d\,\nabla^2 c - g\,\phi_{3D}\Big(\textstyle\sum_i \bar{\phi}_i\Big)\,c, \tag{2}$$

$$\dot{\alpha} = k_\alpha\,\phi_{3D}, \tag{3}$$

with zero flux on the boundary except where the nutrient is held. The last term of Eq. (1) moves the edge of the biofilm up the nutrient gradient with the speed $r\,c/(k+c)$ and is called the front term here. Two details differ from the equations as printed, both confirmed by the authors. The consumption of Eq. (2) is $g\,\phi\,c$ in the paper's notation, $g\,\phi_{3D}c$ here (the printed $g\,\phi$ is a misprint). The front direction is regularised as $\nabla c/(|\nabla c|^2+\varepsilon^2)^{1/2}$ with $\varepsilon^2 = 10^{-6}$ ($\varepsilon^2 = 10^{-8}$ changes the seed stress by 0.1 %; a much larger $\varepsilon^2 = 10^{-4}$ lowers the mixed stress after the antibiotic by 30 %, because ∇c is weak there, Appendix B.3). The factor $\sum_i \bar{\phi}_i$ in Eq. (2) is the coupling term of this work (Section 2.3); in Klempt et al. (2024) the whole biofilm consumes. The mechanical term $\phi_{3D}\,\mu\,(\mathbf{I}:\mathbf{C}_e - 3)$ of their Eq. 30, which they neglect because μ is small against the other parameters, is neglected here as well, so the mechanics is coupled one way. The elastic response is neo-Hookean with the stiffness $E(\phi_{3D}) = E_0\,(\phi_{3D}^2 + f)$, $E_0 = 10$ Pa, $\nu = 0.49$ and a void floor $f = 10^{-3}$ (an assumption that keeps the empty cells solvable; $f \to 0$ changes the seed stress by 0.13 %).

2.2 Point model

At a material point the multi-species model of Klempt et al. (2026) describes the volume fractions ϕ_i and the living fractions ψ_i of N species and the void ϕ_0, with $\sum_i \phi_i + \phi_0 = 1$. It follows from the extended Hamilton principle with the free energy $\Psi = -\frac{1}{2}c^*\,\bar{\boldsymbol{\phi}}^{\mathrm{T}}\mathbf{A}\,\bar{\boldsymbol{\phi}}$, $\bar{\phi}_i = \phi_i\psi_i$, a dissipation with the viscosities η_i, $\eta_{\phi,i}$ and the constraint enforced by the multiplier γ:

$$\begin{aligned}(\eta_{\phi,i} + \eta_i\psi_i^2)\,\dot{\phi}_i + \eta_i\phi_i\psi_i\,\dot{\psi}_i &= c^*\psi_i\textstyle\sum_j A_{ij}\bar{\phi}_j - \gamma - (\text{barrier}),\\ \eta_i\,(\phi_i\psi_i\,\dot{\phi}_i + \phi_i^2\,\dot{\psi}_i) &= c^*\phi_i\textstyle\sum_j A_{ij}\bar{\phi}_j - \alpha^* b_i\,\psi_i - (\text{barrier}),\end{aligned} \tag{4}$$

Here A_{ij} are the interactions, b_i the sensitivity of species i to an antibiotic of concentration α^*, c^* the nutrient, and the barrier terms keep ϕ_i, ψ_i in $(0,1)$. The antibiotic lowers ψ_i through the term $\alpha^* b_i\psi_i$, against the nutrient term $c^*\phi_i\sum_j A_{ij}\bar{\phi}_j$ (Section 2.4). The implementation used here reproduces the fourteen numerical examples of the paper and the figures of Fritsch et al. (2025) within 0.013. Two printed details read differently: Eq. 17 as printed carries a $+\gamma$ that the figures do not show, and the four-species interaction matrix of Eq. 19 reproduces the figures without its prefactor $\frac{1}{2}$. The model is used as published, with its own constraint and its implicit time integration (steps of 10^{-4} in its own time unit, Newton's method). For the examples the synthetic two-species data of Fritsch et al. (2025), case I, are used with their true parameters: $\mathbf{A} = \left(\begin{smallmatrix}1 & 0.1\\ 0.1 & 1\end{smallmatrix}\right)$, $b = (1,2)$, $\eta = (1,2)$, $c_0^* = 100$.

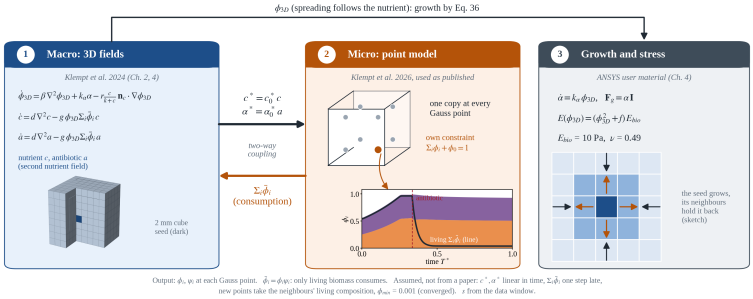


Figure 1: The coupling: the fields ϕ_{3D}, c and a of the growth model, the point model at every integration point with the inputs $c^* = c_0^* c$, $\alpha^* = \alpha_0^* a$ and the output $\sum_i \bar{\phi}_i$, and the mechanics with $\mathbf{F}_g = \alpha \mathbf{I}$.

2.3 Coupling

The two models are connected at every integration point without changing either (Figure 1, Algorithm 1).

1. *Fields to point model.* The point model sees $c^* = c_0^* c$ and $\alpha^* = \alpha_0^* a$, with a the antibiotic, transported by a second field with the diffusivity and uptake of the nutrient, $\dot{a} = d\nabla^2 a - g\,\phi_{3D}(\sum_i \bar{\phi}_i)\,a$. Between the increments t_n and t_{n+1} of the field both are interpolated linearly in time over the inner steps of the point model,

$$c^*(t) = c_0^* \Big[\tfrac{t_{n+1}-t}{\Delta t}\, c_n + \tfrac{t-t_n}{\Delta t}\, c_{n+1} \Big], \qquad t_n \le t \le t_{n+1}. \tag{5}$$

   Klempt et al. (2026) already use a time-dependent c^* (their four-species case 3), so the point model accepts such an input without change.

2. *Point model to fields.* Only the living biomass consumes: the uptake of nutrient and antibiotic is $g\,\phi_{3D}(\sum_i \bar{\phi}_i)$, Eq. (2). The point model normalises its volume, so $\bar{\phi}_i \in [0,1]$ is the living volume fraction of species i at the point and $\phi_{3D} \sum_i \bar{\phi}_i \le \phi_{3D}$ the living volume fraction in the field. The void ϕ_0 has no living fraction and is not part of the sum. A dead biofilm consumes nothing. The growth law (3) is unchanged: α measures the biomass produced, alive or dead.

3. *Time.* Klempt et al. (2024) normalise time with a reference time chosen for the problem at hand. Here $T^* = 1$ is taken as the duration covered by the point model's data, so that both models run on the same time. The point model advances $s\,\Delta T^*$ per increment, with s the length of that window in its own units, not a fitted value ($s = 0.1$ for case I, 1000 steps of 10^{-4}). The antibiotic onset $T^*_{\mathrm{ab}} = 1/3$ of case 4 of Klempt et al. (2026) is taken as the same fraction of the run, not as a duration in the point model's own time.

4. *Start of the point model.* It starts at a point once ϕ_{3D} there reaches $\phi_{\min} = 0.001$. Where a seed is placed it starts from the seed composition. A point that the spreading biofilm reaches later takes the living composition of its colonised neighbours (the neighbours' volume fractions, split in the ratio of their $\bar{\phi}_i$) and starts alive ($\psi_i = 0.999$, $\gamma = 0$). This is because the biofilm spreads through the growth of its living cells. Inheriting the neighbours' living fractions as well gives the same results in the maze (Section 4.4), so what matters is the volume fraction a new point receives, not its living state. The papers do not fix this rule; it is an assumption, and the alternative of starting from the seed composition is compared in Section 4.4.

What this coupling can and cannot change follows from the structure of the equations and is stated next.

Algorithm 1 One increment $t_n \to t_{n+1}$ of the coupled run (user-element form)

Require: at every integration point α_n, the point-model state $(\phi_i, \psi_i, \gamma)_n$ and the pair (c, a) seen in the previous increment; the fields ϕ_{3D}, c, a of t_n
Ensure: displacements, stresses and state of t_{n+1}

repeat ▷ Newton iteration of the equilibrium
  for all integration points **do**
    $\alpha_{n+1} \leftarrow \alpha_n + k_\alpha \, \phi_{3D} \, \Delta t$ ▷ Eq. (3)
    $\mathbf{F}_e \leftarrow \mathbf{F}\,\mathbf{F}_g^{-1}$, $\mathbf{F}_g = \alpha_{n+1}\mathbf{I}$; stress and tangent from $W(\mathbf{F}_e)$ and $E(\phi_{3D})$
    if $\phi_{3D} \geq \phi_{\min}$ or the point model has started **then**
      if not yet advanced in this increment **then**
        (c_n, c_{n+1}), (a_n, a_{n+1}) from the previous and the current value at the point
        n_{sub} steps of $s\Delta t/n_{\text{sub}}$ of Eq. (4) with c^*, α^* from Eq. (5)
        store $(\phi_i, \psi_i, \gamma)_{n+1}$ and $\sum_i \bar{\phi}_i$
      end if
    end if
  end for
until the residual meets the convergence criterion
advance ϕ_{3D} (Eq. (1)), c and a (Eq. (2) with $\sum_i \bar{\phi}_i$)

2.4 What the structure of the coupled equations implies

The main result of this work follows from Eqs. (1) to (4) alone: the composition reaches the growth and the stress along a single path. This bounds what any change of the composition can do to the mechanics, and it says what a model needs if the composition is to change the stress. Three further properties follow from the same equations; all four are used to read the examples.

The path from the composition to the stress. The composition enters the field equations only through the factor $\sum_i \bar{\phi}_i$ in the consumption of Eq. (2). The nutrient enters Eq. (1) only in the front term, through the speed $r\,c/(k+c)$ and the direction $\mathbf{n}_c$, and it does not enter the growth law (3) at all. The composition therefore reaches ϕ_{3D}, α and the stress along one path only,

$$\sum_i \bar{\phi}_i \xrightarrow{(2)} c \xrightarrow{\text{front term of (1)}} \phi_{3D} \xrightarrow{(3)} \alpha \longrightarrow \boldsymbol{\sigma}. \tag{6}$$

It has no effect wherever that path is cut: in the interior of a biofilm, where $\nabla\phi_{3D} = \mathbf{0}$, and wherever the nutrient is uniform, where $\nabla c = \mathbf{0}$ and the front term vanishes. Two bounds follow. Since $0 \leq \phi_{3D} \leq 1$, integrating Eq. (3) gives $\alpha(T) - 1 = k_\alpha \int_0^T \phi_{3D}\,dt \leq k_\alpha T$. In the seed of the examples $\alpha - 1$ reaches about 0.9 of this bound at $T^* = 1$, so no change of the composition can raise the seed growth by more than about 10 %. It can lower it only through the edge of the seed, where $\phi_{3D} < 1$ enters the mean. The front speed $r\,c/(k+c)$ with $k = 1$ and $0 \leq c \leq 1$ lies between 0 and $r/2$, so the composition changes the speed of the edge by at most a factor of two through the nutrient. In practice it changes the speed by the ratio of $c/(1+c)$ at the two nutrient levels (by 37 % at the seed means 0.54 and 0.91 of Section 4.2). The stress is set by the jump of α across the edge of the seed, so the composition acts on the stress through the position and sharpness of that edge, not through the amount of growth. Both bounds follow from the published equations, with a consequence for the modelling: a composition that is to change the stress needs a second path. This can be a growth law in which the rate counts the living biomass, $\dot{\alpha} = k_\alpha \phi_{3D} \sum_i \bar{\phi}_i$, a front speed multiplied by $\sum_i \bar{\phi}_i$, or a stiffness that depends on the species. Each of these is a change of the model (Section 5).

The nutrient adjusts almost at once and depends on the composition through one product. With the constants of Table 1 the nutrient of Eq. (2) adjusts within about $4 \times 10^{-8}\,T^*$,

so at every time it is close to the solution of the linear problem

$$\nabla^2 c = \frac{\Lambda^2}{\ell^2}\,\kappa\, c, \qquad \kappa = \phi_{3D}\textstyle\sum_i \bar{\phi}_i, \qquad \Lambda^2 = \frac{g\ell^2}{d} = 1, \tag{7}$$

with $\ell = 1$ mm the half edge of the cube and Λ the Thiele modulus of the conversion. Three consequences follow. The nutrient is proportional to its held value, so the examples do not depend on the level of the supply. It depends on the composition only through the living volume fraction κ of the field, not on which species is alive. It is also depleted over the length $\lambda = \ell/(\Lambda\sqrt{\kappa})$, which is 1 mm, the half edge of the cube, for fully living biofilm and 2.6 mm when $\sum_i \bar{\phi}_i = 0.15$. With $\Lambda = 1$ the nutrient is never exhausted inside the biofilm; it decreases by a factor of order one across the cube. A slab of living biofilm with uniform κ between the supply face and the opposite wall gives $c = \cosh((\ell - y)/\lambda)/\cosh(2\ell/\lambda)$ and at the centre 0.41 for $\kappa = 1$ and 0.82 for $\kappa = 0.15$. The seed means of Section 4.2 (0.54 and 0.91) lie above these slab values because the biofilm outside the seed has $\phi_{3D} < 1$ and does not fill the cube. The antibiotic field obeys the same equation with the same κ, so it is depleted over the same length; where living biofilm shields the nutrient it shields the antibiotic as well.

The antibiotic acts through a balance that depends on the density of the species. The living fraction of species i evolves by the second line of Eq. (4), Eq. 17 of Klempt et al. (2026). At fixed ϕ_i and away from the barriers,

$$\eta_i\,\phi_i^2\,\dot{\psi}_i = c^*\phi_i \textstyle\sum_j A_{ij}\phi_j\psi_j - \alpha^* b_i\,\psi_i, \tag{8}$$

so ψ_i rises where the nutrient term exceeds the antibiotic term and falls where it does not. For a species that dominates its point ($A_{ii}\phi_i^2\psi_i \gg \sum_{j\neq i} A_{ij}\phi_j\psi_j$) the balance is $c^* A_{ii}\phi_i^2$ against $\alpha^* b_i$, and the relative rate is

$$\frac{\dot{\psi}_i}{\psi_i} = \frac{c^* A_{ii}}{\eta_i} - \frac{\alpha^* b_i}{\eta_i\,\phi_i^2}. \tag{9}$$

The nutrient term gives a rate that does not depend on how much of the species is present, the antibiotic term a decay rate that grows as $1/\phi_i^2$. The antibiotic kills slowly a species that fills its point, and almost at once the same species present with a small volume fraction. The rate changes sign on the line $c^* A_{ii}\phi_i^2 = \alpha^* b_i$, but ψ_i decays exponentially rather than reaching zero, and ϕ_i evolves at the same time by the first line of Eq. (4). The equations therefore give a balance with a graded outcome, not a sharp boundary between living and dead regions. In the examples $c^* = c_0^* c$ and $\alpha^* = \alpha_0^* a$ with $c_0^* = \alpha_0^*$, so the balance for species 1 ($b_1 = 1$) reads $c\,\phi_1^2$ against a. It is approached only where the nutrient is as high as the antibiotic and the species fills its point, next to the supply face. Everywhere else the species declines at a rate set by $a/\phi_1^2 - c$. Where the nutrient is high the decline is slow, where it is low it is fast; this is the spatial pattern of Section 4.2. The same expression says that a mixed seed, in which each species holds about half the volume, loses its living fraction much faster than a seed of one species. It also says that the state given to newly reached material decides its fate under the antibiotic (Section 4.4).

The time link is not a parameter of the model. Both lines of Eq. (4) are linear in the rates, and every rate carries the viscosity of its species. Running the point model on the time T^* of the field with $t = s\,T^*$ therefore equals running it on its own time with the viscosities η_i/s and $\eta_{\phi,i}/s$ (the barrier penalty $K_{p,i} = 10^{-4}\eta_i$ scales with them). With $T^* = 1$ taken as the duration of the data the point model was calibrated on, s is fixed by the data and the viscosities keep the values of the calibration; no new constant enters.

2.5 A second path: growth of the living biomass

The second part of the results (Section 4.6) changes one term of the model, derived from the energy, to open a second path from the composition to the growth. In the energy of Klempt et al.

Table 1: Constants of the examples. Conversion: Table 2 of Klempt et al. (2024) on the 2 mm cube with the half edge $\ell = 1$ mm, so that $g\ell^2/d = 1$.

quantity	symbol	value	source
diffusion of ϕ_{3D}	β	0.02 mm$^2/T^*$	Klempt et al. (2024), converted (assumption)
front speed, half saturation	r, k	10 mm/T^*, 1	Klempt et al. (2024), converted ($r = 100$ printed in T^{*-1}, read as μm/T^* on the 20 μm cube)
growth rate	k_α	$10^{-3}\,T^{*-1}$	Klempt et al. (2024)
nutrient diffusion, uptake	d, g	10^8 mm$^2/T^*$, $10^8\,T^{*-1}$	Klempt et al. (2024), converted
antibiotic diffusion, uptake	d_a, g_a	$= d$, $= g$	assumption
stiffness, Poisson ratio	E, ν	10 Pa, 0.49	Klempt et al. (2024)
void floor	f	10^{-3}	assumption
point model	$\mathbf{A}$, b, η, c_0^*	case I	Fritsch et al. (2025)
antibiotic level, onset	α_0^*, T_{ab}^*	100, 1/3	Klempt et al. (2026), case 4
time link	s	0.1	from the data of case I
start threshold	$\phi_{\min}$	0.001	assumption (converged)
front regularisation	ε^2	10^{-6}	assumption

(2024), their Eq. 20, the term $-\bar{k}_\alpha\,\alpha\,\phi$ couples the growth variable to the biofilm. Its variation gives the source $\bar{k}_\alpha\alpha$ in the evolution of ϕ (their Eq. 30) and the driving force $\bar{k}_\alpha\phi$ of α (their Eq. 32), so that every part of the biofilm, living or dead, grows. Growth of the active biomass only is the standard assumption of multi-species biofilm models since Wanner and Gujer (1986). Here the term becomes

$$-\bar{k}_\alpha\,\alpha\,\phi_{3D}\textstyle\sum_i \bar{\phi}_i, \tag{10}$$

with $\sum_i \bar{\phi}_i$ given at each point by the point model and not varied, in the same way as Klempt et al. (2024) treat their growth term h_ϕ^* as a property of the material that does not adjust to make the functional stationary (their Remark 1). The variations with respect to ϕ_{3D} and α then give, after division by the viscosities,

$$\dot{\phi_{3D}} = \beta\,\nabla^2\phi_{3D} + k_\alpha\,\alpha\textstyle\sum_i \bar{\phi}_i - r\,\frac{c}{k+c}\,\mathbf{n}_c\cdot\nabla\phi_{3D}, \tag{11}$$

$$\dot{\alpha} = k_\alpha\,\phi_{3D}\textstyle\sum_i \bar{\phi}_i, \tag{12}$$

in place of Eqs. (1) and (3); Eq. (2), the point model and the coupling are unchanged. Where the point model has not started, $\sum_i \bar{\phi}_i = 0$ and the biofilm does not grow locally. The bound of Section 2.4 becomes $\alpha(T) - 1 \le k_\alpha \int_0^T \sum_i \bar{\phi}_i\,dt$, so the antibiotic now acts on the growth directly. A second form, in which $\sum_i \bar{\phi}_i$ divides the viscosity of α instead of multiplying the energy, gives Eq. (12) with Eq. (1) unchanged. The two forms agree within 0.2 % in the growth and 0.4 % in the seed stress on every grid of Section 4.6, because the source $k_\alpha\alpha$ of Eq. (1) is small against the front term. The form (10) is used because it follows from the energy.

2.6 Parameters

All constants are those of the papers (Table 1): Table 2 of Klempt et al. (2024), converted from their 20 μm cube to the 2 mm cube of the examples, and the antibiotic of case 4 of Klempt et al. (2026), $\alpha_0^* = 100$ from $T^* = 1/3$. With $d = 10^8$ mm$^2/T^*$ the fields adjust within about $4 \times 10^{-8}\,T^*$, far faster than the growth. A diffusion slower by eight orders of magnitude ($d = 1$ mm$^2/T^*$) is only a sensitivity: the antibiotic then reaches the seed more slowly and kills markedly less of it. Values not given by the papers are marked as assumptions: the length and time scale of the conversion, $\phi_{\min}$ (results change by less than 1 % below 0.001), the start rule of new points, the void floor f and the regularisation ε.

3 Implementations and verification

The coupled model is solved in three codes that share the equations, the parameters and the compiled point-model routine, but not the discretisation. The first is a finite-volume code (7-point Laplacian, second-order upwind front term with the minmod limiter, RK4 in time, hexahedral mechanics with selective reduced integration). The second is the neighboured-element user element of the original authors' group in ANSYS (Blaszczyk et al., 2022; Rudolf et al., 2025), in which the Laplacian and the gradients are fitted on the Gauss points from 30 neighbours by weighted least squares and the fields are updated explicitly. The third is a Galerkin user element in Abaqus (8-node hexahedra, SUPG (Brooks and Hughes, 1982) for the front term, $\bar{\mathbf{B}}$ mechanics, implicit Euler with Newton's method). Eq. (2) and the antibiotic field are integrated with their time derivative by the backward Euler method, one linear solve per field step at the cost of the problem without it; the method is stable for any step, whereas an explicit update with the diffusivity of Table 1 would need steps below $10^{-8}\,T^*$. With $d\,\Delta t \gg \ell^2$ the term $(c^{n+1}-c^n)/(d\,\Delta t)$ is below 10^{-6} of the others. The finite-volume and Abaqus results reported here were computed in the limit $d\,\Delta t \to \infty$ of this step; with the term the finite-volume code changes the seed living fraction, c, $\alpha - 1$ and the von Mises stress by at most 0.1 % (32^3, $\Delta t = 0.0025$). Three numerical details of the neighboured element were corrected before use, without changing its equations (Appendix A.1). Differences between the codes therefore come from the fields and the mechanics only.

The verification follows the usual steps (Roache, 2002; Celik et al., 2008); Appendix B gives the details. With uniform fields every code reproduces the point model alone to round-off (2×10^{-14} in the finite-volume code, 1.8×10^{-12} in the element), so the hand-over of $\sum_i \bar{\phi}_i$ to the consumption is exact. With a manufactured solution the finite-volume and the Galerkin code converge at second order in L_2 for ϕ_{3D} and c (observed orders 1.7 to 2.05; the SUPG form of ϕ_{3D} at about 1.5). The analytic tangent of the Galerkin element agrees with central differences to 5×10^{-11}. The neighboured element does not converge next to the walls (Appendix A.1). The mechanics of two codes agree to 1 % on the same growth field, so differences in the stress between the codes come from the growth field.

One numerical property of the model sets the accuracy of the stress and is therefore stated here. The front term of Eq. (1) is an advection with the speed $r\,c/(k+c)$ along $\mathbf{n}_c$ against the small diffusion β: the cell Péclet number $rh/(2\beta)$ is about 31 on a 16^3 grid of the 2 mm cube. A central discretisation therefore oscillates and needs stabilisation. A constant added diffusion $rh/4$, which keeps the element as delivered stable, smears the edge of the seed over a few cells. The seed stress, which is set by the jump of α across that edge, follows the smearing and is about a quarter of the converged value on 16^3 and still rising on 96^3. The growth variable converges with every treatment. The three second-order treatments (upwind with the minmod limiter, SUPG, upwind-biased least-squares gradients) give the same seed stress within a few per cent from 32^3 on, and the three codes agree within their grid-convergence bands (Appendix B.2). All results below use a second-order treatment.

The examples were then run on 32^3, 64^3, 128^3 and 192^3 cells, with the outer step 0.0025 (the step study is in Appendix B.3). The three-grid index of Celik et al. (2008) with the safety factor 1.25 is taken on the three finest grids (Figure 2, Table 5). The growth converges with a GCI of 0.4 to 4.3 % in all examples, the seed stress with 0.1 % without and 1.4 % with the antibiotic when the species start apart. When they start mixed the stress converges slowly (observed order 0.4 to 1.0, GCI 19 to 41 %; the cell Péclet number of Appendix B.4), because the front carries intermediate values of ϕ_{3D} at a nearly discontinuous edge. The largest ϕ_{3D} in the domain still rises with the grid (0.60, 0.69, 0.73, 0.75), though by less at each refinement. No run on 32^3 to 192^3 left $[0, 1]$ by more than 2×10^{-4}. Part of the slow convergence is the numerical diffusion of the minmod limiter: with the less diffusive superbee limiter the mixed seed stress with antibiotic converges with an observed order of 1.2 to 1.3 and a GCI of 4.6 to 6.3 % to 3.8 to 3.9×10^{-4} Pa (Appendix B.4).

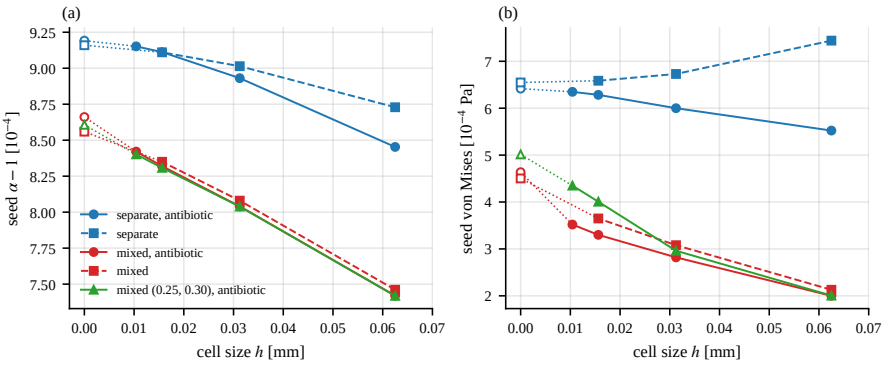


Figure 2: Grid convergence of the two-species examples (finite-volume code, outer step 0.0025; Table 5): seed means of (a) $\alpha - 1$ and (b) the von Mises stress at $T^* = 1$ against the grid. Open symbols: Richardson extrapolation from the three finest grids.

4 Results

4.1 Set-up

The examples use a 2 mm cube with a cross-shaped seed of $\phi_{3D} = 1$ at its centre (32 elements of 0.25 mm, 0.5 mm^3), the nutrient held at $c = 1$ in the layer of elements at the face $y = -1$ mm and the antibiotic held at $a = 1$ in the outer layer of elements on all faces from $T^* = 1/3$, with no flux elsewhere. Species A is $i = 1$, species B $i = 2$. In the separate example species A fills the seed and species B four small blocks of $0.25 \times 0.25 \times 1.0$ mm along the four edges parallel to z, at the height of the seed; in the mixed example both start in the seed, in equal amounts ($\phi_1 = \phi_2 = 0.275$) or in the proportions of Fritsch et al. (2025) (0.25, 0.30). The front term is on in every run. The values below are seed means at $T^* = 1$ from the Python code (TVD, newly reached points inherit the living composition of their neighbours), which reaches the finest grids. The element gives the same composition within its grid error and its growth within 4 % after extrapolation. The stress is solved on 16^3 hexahedra with the element's integration, with ϕ_{3D} and α averaged over the field cells of each element. Solving the same 64^3 field on 32^3 and 64^3 elements changes the seed von Mises stress by 0.2 %, so the grid dependence of the stress comes from the field. Stresses are given in Pa for $E_0 = 10$ Pa; since the mechanics is coupled one way they scale with E_0, and σ/E_0 follows by dividing by 10 Pa. The colour scale of the hydrostatic stress in Table 3 of Klempt et al. (2024) spans -6.5 to $+3 \times 10^{-4}$, labelled MPa; with the shear modulus of their Table 2 given in Pa, these values are in Pa, the same order as here.

4.2 Composition and living biomass

Table 2 gives the seed means and Figure 3 the fields of the separate and the mixed example with antibiotic. Without antibiotic the living volume fraction of the seed stays near 0.96; the antibiotic lowers it to 0.15 when the species start apart and to 0.04 when they start mixed. Species B is removed first: its nutrient term $c^* A_{22}/\eta_2$ is half that of A, while the antibiotic term per viscosity, $\alpha^* b_i/\eta_i$, is the same for both. The nutrient field decides where the antibiotic acts, as Eq. (9) says: species A declines at the rate $(a/\phi_1^2 - c)\, c_0^* s/\eta_1$ per unit of T^* ($\eta_1 = 1$), so where the nutrient is high it survives, and where it is low it dies. The seed values give the size of the rates. In the separate example the seed holds $\phi_1 = 0.91$ with $c = 0.91$ and $a = 0.98$ at $T^* = 1$, a decline of 2.7 per unit of T^*. This is consistent with the fall of ψ_1 from 0.98 to 0.16 between the onset and $T^* = 1$. In the mixed seed species A holds only $\phi_1 = 0.57$, the decline is 21 per unit of T^*, and its living fraction is gone within $0.2\,T^*$ of the onset. The difference between 0.15 and 0.04 therefore comes from the balance in the point model, not from the field: the same species at the same nutrient and antibiotic dies about eight times faster when it shares its point. When the

Table 2: Seed means at $T^* = 1$, Python, 64^3 cells, outer step 0.025. At the step 0.0025 the composition changes by less than 3 % and the stress with antibiotic by +6 % (separate) and −13 % (mixed) (Appendix B.3); grid-converged growth and stress at the step 0.0025 are in Table 5.

example	$\sum_i \bar{\phi}_i$	$\phi_1/(\phi_1+\phi_2)$	c	$\alpha - 1$	von Mises [Pa]
separate	0.965	0.978	0.536	9.06×10^{-4}	6.56×10^{-4}
separate, antibiotic	0.152	0.951	0.911	8.95×10^{-4}	5.68×10^{-4}
mixed, equal amounts	0.961	0.859	0.608	8.33×10^{-4}	3.50×10^{-4}
mixed, equal amounts, antibiotic	0.038	0.610	0.973	8.30×10^{-4}	3.23×10^{-4}
mixed (0.25, 0.30), antibiotic	0.037	0.510	0.973	8.30×10^{-4}	3.22×10^{-4}

biomass dies the nutrient is no longer consumed and rises in the seed, from 0.54 to 0.91 in the separate example, as the depletion length of Eq. (7) grows from 1 to 2.6 mm. The antibiotic, which obeys the same equation, reaches 0.98 in the seed. Living volume fraction, shares and nutrient change by less than 3 % between the grids (32^3 to 128^3) and the outer steps (0.025 to 0.0025) used.

Along the line $x = z = 0$ through the seed (Figure 4) the living volume fraction stays high near the supply face after the antibiotic in the separate example and falls with the distance from it, while in the mixed example it falls everywhere within about $0.1\,T^*$ of the onset. The front fills the line from both ends by $T^* \approx 0.15$; in the separate example a band of lower ϕ_{3D} then travels from $y = 1$ mm towards the supply face and closes by $T^* \approx 0.55$.

4.3 Antibiotic over time

With the same total dose given constantly, in pulses or as a sine from $T^* = 1/3$ (separate example, 64^3; the time series are in the supplementary material), $\alpha - 1$ changes by 0.1 % and the seed stress by 0.8 % between the three schedules. The living volume fraction of the seed, in contrast, follows the dose. In the pauses of the pulsed dose it recovers from 0.17 to up to 0.34 and ends at 0.30. The sine gives a lower minimum (0.11 against 0.15) and a higher end value (0.19) than the constant dose (0.15). How the dose is given acts on the composition and the living biomass, hardly on the mechanics.

4.4 Which composition does new biofilm receive?

The rule for newly reached points (Section 2.3, item 4) is not fixed by the papers. The point model has no transport term, and Eq. (1) moves ϕ_{3D} without carrying the point state, so the composition travels with the biofilm only through the state that a newly reached point is given. The two rules compared here differ in the volume fraction they assign. Inheriting the neighbours' amounts in the proportions of their living biomass gives a new point a volume fraction close to that of the established biofilm behind it. The start state of the colony gives it the small start fractions of the colony (0.25 to 0.3 per species). By Eq. (9) the antibiotic term scales as $1/\phi_i^2$, so a point started from the start state loses its living fraction at a rate about 13 times that of an inherited point (a/ϕ_1^2 of 16 against about 1.2). Where the biofilm has stopped spreading the rule hardly matters. With the start values of the seed instead of the living composition of the neighbours, the seed values of the cube examples change by a few per cent ($\sum_i \bar{\phi}_i$ 0.147 against 0.152 in the separate example with antibiotic). In the maze of Klempt et al. (2024, Fig. 11), where the biofilm keeps spreading after the antibiotic and most of the colonised area at the end was reached after the onset, the rule decides the outcome (Figure 5, set-up in Section 4.8). The living volume fraction of the colonised region falls to 0.10 shortly after the onset of the antibiotic at $T^* = 20$. It recovers to 0.35 at $T^* = 60$ when new cells take the state of their neighbours, because the biofilm keeps spreading and every newly reached cell starts with a large volume fraction, so the antibiotic, whose term scales as $1/\phi_i^2$, removes it slowly. Inheriting the

neighbours' living fractions too, instead of starting the new cell alive, gives the same recovery (0.347 against 0.346 at $T^* = 60$): the recovery comes from the inherited volume fraction, not from the living state. It stays at 0.04 when they start from the start state of the colony, because thinly started cells die at once. The Galerkin element gives the same with the start-state rule (0.04 to 0.07).

4.5 What reaches the stress

The growth $\alpha - 1$ hardly depends on the composition, as the bound of Section 2.4 requires: Eq. (3) is unchanged, and the antibiotic lowers ψ_i but not ϕ_{3D}. The seed growth is within 10 % of the bound $k_\alpha T^*$ when the species start apart and within 15 % when they start mixed. Extrapolated (Table 5), it is 9.17 without and 9.19×10^{-4} with the antibiotic in the separate example, 8.74 and 8.71×10^{-4} in the mixed one, differences of 0.2 and 0.3 %. The composition reaches the stress along the path (6) only, through the front speed, which the nutrient rise from 0.54 to 0.91 raises by 37 % at the edge of the seed. When the species start apart, no effect of the antibiotic on the converged seed stress is resolved: 6.57 without and 6.42×10^{-4} Pa with the antibiotic (extrapolated, GCI 0.1 and 1.4 %). The difference of 2 % is about the two bands together. When they start mixed, the superbee series extrapolated from 128^3, 192^3 and 256^3 gives 4.26 without and 3.95×10^{-4} Pa with the antibiotic (GCI 2.4 and 6.3 %), 7 % lower, about the two bands together (9 %); with minmod the difference is hidden in bands of 31 and 41 %. The antibiotic may lower the mixed stress by a few per cent, but this is not resolved on these grids. Only extrapolated values can be compared: on a fixed grid the difference is larger (6.73 against 6.00×10^{-4} Pa on 64^3) because the two cases converge from opposite sides. The mixed seed carries less stress than the separate one, because its edge is less sharp: the front carries intermediate values of ϕ_{3D}. With the antibiotic the extrapolated values are 3.95 (superbee, 128^3 to 256^3, GCI 6.3 %) against 6.42×10^{-4} Pa (GCI 1.4 %), a ratio of about 0.6; with minmod the mixed value is only known within 41 % (Table 5). Beyond the seed, the extent of the biofilm is not changed at the full front speed, which fills the cube in every example. At one tenth of it the antibiotic enlarges the colonised volume (0.27 against 0.23, Table 4), because dead biofilm keeps spreading. A limitation follows from the same structure: the front speed $r\,c/(k + c)$ depends on the nutrient only, so dead biofilm keeps spreading after the antibiotic. In the maze about two thirds of the cells reached after the antibiotic hold only dead biofilm when new cells start from the start state. Tying the front speed or the growth rate to the living biomass would open a second path and changes the model (Section 5).

Figure 6 shows the hydrostatic stress at $T^* = 1$ in the form of Fig. 8 of Klempt et al. (2024). The seed, which grew longest, is in compression (down to -2×10^{-3} Pa in the separate example, most in its half towards the supply face), and the biofilm around it, which arrived later and grew less, is in tension, most along the walls. The mixed example has the same pattern with a smoother edge.

What sets the seed stress. A uniform growth swells freely and gives no stress, so the seed stress measures how unevenly the stiff biofilm grows. To test this, 51 runs on 64^3 were compared at $T^* = 1$ (Figure 7): the four examples and the slower front of Section 4.7, a nutrient held as $0.5 + 0.5 \sin(500\, s\, T^*)$ or at its mean 0.5, and sixteen pairs of α_0^* (10 to 300) and $g\ell^2/d$ (0.5 to 2) in the separate example with antibiotic, each with the published growth law and the law of Section 2.5, and the antibiotic schedules of Section 4.3 with the latter. The mean of $(\phi_{3D}^2 + f)\,|\nabla\alpha|$ over the seed and the shell of one element around it, the gradient of the growth weighted with the stiffness, correlates with the seed stress with 0.94 (published law, 26 runs) and 0.90 (living-biomass law, 25 runs). The growth of the seed itself does not (-0.03 for the published law). With the full front speed the cube is full at $T^* = 1$ and ϕ_{3D} is the same in the seed and around it. The difference of α remains, because α records when the biofilm arrived: the seed grows from the start, its surroundings only once the front has reached them. A lower

Table 3: Growth of the living biomass, term (10), against the published law: seed means at $T^* = 1$ on 128^3 cells (outer step 0.0025), the ratio to the published law on the same grid, and the extrapolated growth with its GCI (32^3, 64^3, 128^3). Growth in 10^{-4}, von Mises stress in 10^{-4} Pa.

example	$\alpha - 1$	ratio	von Mises	ratio	$\alpha - 1$ extrap.	GCI	$\sum_i \bar{\phi}_i$
separate	8.31	0.91	6.12	0.93	8.36	0.7 %	0.97
separate, antibiotic	4.69	0.51	5.25	0.84	4.74	1.3 %	0.16
mixed	7.24	0.87	3.27	0.90	7.43	3.3 %	0.96
mixed, antibiotic	2.87	0.34	2.31	0.70	2.88	0.5 %	0.04

nutrient slows the front and enlarges this difference. Holding the nutrient at 0.5 instead of 1 raises the seed stress by 31 % and lowers the seed growth by 9 %; the sine with the same mean gives the same within about 1 %, so at this frequency only the mean supply matters. The slower front also leaves a jump of ϕ_{3D} (about 0.33) and gives the highest stresses. The mixed seed carries less stress at a similar gradient of α, because the front carries intermediate values of ϕ_{3D} and the stiffness around the seed is lower. These are values on one grid, and the stress of the mixed example converges slowly (Table 5).

4.6 Growth of the living biomass

With the term (10) the antibiotic acts on the growth (Table 3). On 128^3 cells (outer step 0.0025, Eq. (2) with its time derivative) it lowers the seed growth to 0.51 of that of the published law in the separate example and to 0.34 in the mixed one; without the antibiotic the ratios are 0.91 and 0.87. The time averages of $\sum_i \bar{\phi}_i$ in the seed over $[0, T^*]$ of the published runs had suggested 0.54 and 0.30 with the antibiotic. The growth converges with a GCI of 0.5 to 3.3 % (32^3, 64^3, 128^3). The seed stress falls by less than the growth: to 0.84 of the published law in the separate example and to 0.70 in the mixed one with the antibiotic, and to 0.93 and 0.90 without it. These ratios are taken on the same grid, because the stresses of these three grids are not yet in the asymptotic range: in three of the four examples they still rise by 7 to 20 % from 64^3 to 128^3 (GCI 14 to 33 %); the separate example without antibiotic converges (GCI 0.7 %). The cause is the resolution of the edge that the front term moves (Appendix B.4). With this one change of the energy the composition reaches the growth and the stress, as the bound predicts, while the front speed, the stiffness and the hand-over of c^*, α^* and $\sum_i \bar{\phi}_i$ stay as published.

4.7 A slower front

With the front speed $r = 10$ mm/T^* of Klempt et al. (2024), converted to the 2 mm cube, the biofilm reaches the walls before the antibiotic starts and fills the cube by $T^* = 1$: 92 to 95 % of the cells hold $\phi_{3D} \geq 0.5$ at the end. To check that the results above do not depend on the cube being full, the four examples were repeated with one tenth of that speed, $r = 1$ mm/T^*, all else unchanged (64^3, outer step 0.025; Table 4). The biofilm then occupies 20 to 27 % of the cube at $T^* = 1$ and is still spreading. The seed means change in the way the path of Eq. (6) prescribes. The nutrient in the seed is higher without antibiotic (0.75 against 0.54 and 0.61), because less living biofilm lies between the supply face and the seed. The antibiotic balance of Eq. (9) therefore tips less far in the separate seed, which keeps a living volume fraction of 0.30 instead of 0.15. The mixed seed, whose species share their points, loses its living biomass as before (0.04). The higher nutrient also lets species A take over the mixed seed further (0.98 against 0.86 without antibiotic). The growth $\alpha - 1$ of the seed stays within 8 % of its value at the full speed, as the bound $k_\alpha T^*$ requires. The antibiotic changes it by 3 % (separate) and 0.2 % (mixed) at the slower speed, against 2 % or less at the full speed. The seed stress is 1.6 to 3.3 times higher at the slower front, because the edge of the seed stays sharper when it moves less

Table 4: One tenth of the front speed ($r = 1$ mm/T^*) against the speed of the paper ($r = 10$ mm/T^*): seed means at $T^* = 1$ and the colonised fraction of the cube ($\phi_{3D} \geq 0.5$), Python, 64^3 cells, outer step 0.025; mixed: equal amounts; r in mm/T^*.

example	r	$\sum_i \bar{\phi}_i$	$\phi_1/(\phi_1+\phi_2)$	c	$\alpha - 1$	von Mises [Pa]	colonised
separate	10	0.965	0.978	0.536	9.06×10^{-4}	6.56×10^{-4}	0.95
separate	1	0.969	0.981	0.745	8.46×10^{-4}	1.04×10^{-3}	0.23
separate, antibiotic	10	0.152	0.951	0.911	8.95×10^{-4}	5.68×10^{-4}	0.95
separate, antibiotic	1	0.299	0.965	0.918	8.70×10^{-4}	1.19×10^{-3}	0.27
mixed	10	0.961	0.859	0.608	8.33×10^{-4}	3.50×10^{-4}	0.92
mixed	1	0.969	0.981	0.756	9.03×10^{-4}	1.14×10^{-3}	0.22
mixed, antibiotic	10	0.038	0.610	0.973	8.30×10^{-4}	3.23×10^{-4}	0.92
mixed, antibiotic	1	0.039	0.658	0.984	9.01×10^{-4}	1.08×10^{-3}	0.20

far. On this grid the antibiotic changes it by 15 % (separate, upwards) and 6 % (mixed). This is the same order as the 16 and 8 % on the same grid at the full speed (relative to the value with antibiotic), where the extrapolation of Table 5 reduces the difference to the grid uncertainty. A grid series at the slower speed was not run. The front speed sets the extent of the biofilm, the seed stress and how far the antibiotic tips the balance; the effect of the composition on the growth does not depend on it.

4.8 Geometry: rigid bars and a maze

Klempt et al. (2024) also grow the biofilm against rigid obstacles (their Section 4.3) and through a maze (Section 4.4). Both were repeated with the coupled model in the Python code, without antibiotic unless stated (Figure 8).

Rigid bars. Four bars after Fig. 9 of the paper, with positions read off the sketch: two along x and two along z, each 0.25 mm square, between 0.5 and 0.75 mm from the centre. They hold no biofilm and no nutrient, and the biofilm slides along them without friction. All other settings are those of Section 4.1 (32^3, outer step 0.0025). The biofilm grows around the bars and closes behind them, and by $T^* = 0.2$ it fills the cube between them. The composition of the seed hardly changes. The bars restrain the swelling: the seed von Mises stress is 7.1×10^{-3} Pa against 7.4×10^{-4} Pa without bars in the separate example, about ten times higher, and 9.5×10^{-4} against 2.1×10^{-4} Pa in the mixed one. The growth $\alpha - 1$ of the seed is 9 and 21 % lower. A grid series with antibiotic (outer step 0.0025) shows that this stress converges slowly: the separate seed gives 2.52, 3.73, 5.04 and 5.40×10^{-3} Pa on 32^3 to 192^3 (+48, +35 and +7 %; observed order 1.3, extrapolated 5.9×10^{-3} Pa, GCI 12 %), the mixed seed 0.93, 1.61, 2.21 and 2.46×10^{-3} Pa (order 0.6, GCI 43 %). With the superbee limiter the series converges faster: 2.69, 4.21, 5.46 and 5.70×10^{-3} Pa in the separate seed (order 1.9, extrapolated 5.90×10^{-3} Pa, GCI 4.5 %), the same limit as with minmod, and 1.13, 1.98, 2.49 and 2.61×10^{-3} Pa in the mixed seed (order 1.7, extrapolated 2.72×10^{-3} Pa, GCI 5.3 %). The seed growth changes by 5 % or less per refinement. The stress is solved on the same 16^3 hexahedra in every run, so the rise does not come from resolving the edges of the bars in the mechanics: a stress mesh of 32^3 changes the separate seed stress by 3 % only (3.61 and 4.87×10^{-3} Pa on 64^3 and 128^3 fields), while the field grid changes it by 35 %. The rise comes from the fields ϕ_{3D} and α next to the bars, which a finer field grid still changes. Against the runs without bars on the same grid the factor grows from about 5 on 32^3 to 8 (separate) and 7 (mixed) on 128^3. The bars therefore raise the seed stress by this factor or more; with the superbee limiter its value with bars is known within about 5 % in both examples.

Asymmetric starts. The examples above are mirror symmetric in x and z. Two variants break this symmetry in the separate example with antibiotic (64^3, full domain, outer step 0.025): species B only in the block along the edge $x > 0.75$, $y < -0.75$ mm with the seed moved by

one element (0.25 mm) towards $-x$, and species B only in one cube of 0.25 mm at the corner $x > 0.75$, $y < -0.75$, $z > 0.75$ mm. On the full domain the symmetric example reproduces the quarter domain to seven digits (32^3). The composition of the seed hardly changes ($\sum_i \bar{\phi}_i = 0.156$ and 0.158 against 0.152, share of A 0.952 against 0.951), nor does its nutrient (0.92 against 0.91). The growth of the seed is 8 and 7 % lower (8.22 and 8.34 against 8.95×10^{-4}), and the seed von Mises stress is 31 and 40 % lower (3.94 and 3.41 against 5.69×10^{-4} Pa). Where species B starts thus changes the stress of the seed much more than its composition. On 128^3 the gap is smaller: the seed stress is 4.87 and 4.29×10^{-4} Pa against 5.90×10^{-4} Pa, 17 and 27 % lower. The asymmetric stresses rise by 22 to 26 % from 64^3 to 128^3, the symmetric one by 4 % (GCI 2.3 % on 64^3 to 192^3), so the size of the gap is not yet grid converged; that it is there on both grids is. The cause has not been analysed further.

Maze. The maze of Fig. 11 of the paper, read off the figure (560 μm across), is run in two dimensions with 4 μm cells and the constants of Table 6 of the paper ($\beta = 10\ \mu\text{m}^2/T^*$, $r = 200\ \mu\text{m}/T^*$, $k_\alpha = 10^{-6}\, T^{*-1}$). The nutrient is held in the centre, the colony of both species starts at the entrance, and the point model is that of the two-species examples. With the antibiotic, $a = 1$ is held at the entrance from $T^* = 20$. The model is two-dimensional, so it gives no stress. The biofilm follows the corridors, also into dead ends, and approaches the centre: at $T^* = 60$ the nearest point with $\phi_{3D} \geq 0.5$ is about 40 μm from the centre with and without antibiotic. A grid series with 4, 2 and 1 μm cells gives at $T^* = 20$ a reach of 71, 68 and 67 μm and colonised areas within 6 %; at $T^* = 60$ the distance is 39 to 43 μm in all runs except the one with antibiotic on 4 μm cells (133 μm), where a single cell decides it, while its area and living volume fraction agree with the finer cells. The living volume fraction of the colonised region is 0.92 without antibiotic; with it, the rule for new cells decides whether the biofilm recovers (Section 4.4).

4.9 A molar crown geometry

The cube examples place the seed in a box. To show the coupling on a real geometry, the same model and constants were run around the crown of a human molar from the Open-Full-Jaw dataset (Gholamalizadeh et al., 2022) (Figure 9). The crown ($6.7 \times 6.8 \times 5.5$ mm) stands with its cervical base on the bottom face of a 10 mm box and is rigid for the fields (no biofilm and no nutrient inside, zero flux through its surface) and for the stress ($\mathbf{u} = \mathbf{0}$). Nutrient and, from $T^* = 1/3$, antibiotic are held at 1 on the five faces around the crown (the saliva side); the bottom face is zero flux (the root side). The biofilm starts as a cervical ring 0.25 mm thick and 1 mm high, species A on the half $x < 0$ and B on $x > 0$. These choices are assumptions of this work. The grid has 80^3 cells ($h = 0.125$ mm; 40^3 for comparison), the outer step is 0.025, the film is taken as the cells with $\phi_{3D} > 0.5$, and the stress is solved on the field grid with the film surface free.

With the paper's front speed the biofilm spreads along the bottom face and outwards and reaches the side faces by $T^* = 1$. The film that covers the bottom face around the crown at the end follows from that speed, which fills the 2 mm cube in the same time (Section 4.7); the composition plays no part in it. The composition is carried with it: the A and B halves stay separate (film mean share 0.50; 0.956 in the A ring, 0.051 in the B ring). The antibiotic enters with the nutrient ($a = c$ in the film, because both are held on the same faces with the same diffusion and uptake; 0.73 on average, 0.24 at the ring). The balance $c\,\phi_1^2$ against a can therefore hold only where a species fills its point. It acts first on B, as $b_2 = 2$ and Eq. (9) require: the living fraction of the B ring falls from 0.71 at the onset to 0.13 at $T^* = 1$, that of the A ring from 0.81 to 0.57. The film mean is 0.37 (0.95 without antibiotic). Growth and stress hardly depend on the antibiotic, as on the cube (film mean $\alpha - 1$ of 4.3×10^{-4} with and without; von Mises 3.0 and 3.2 mPa, with the largest values in the corners between the film, the crown and the bottom face). Between 40^3 and 80^3 the film means change by 3 % and the film stress by 7 %, most of

which is the stress mesh. The example demonstrates the coupling on a real surface geometry, not a clinical treatment. The ring is two cells thick, the crown is rigid and enamel and saliva are not modelled, and the stress of the cervical region with a growth law is not treated here.

5 Discussion

What the coupling shows. The examples confirm what the structure of the equations prescribes (Section 2.4). The composition reaches the growth field only through the consumption $g\,\phi_{3D}(\sum_i \bar{\phi}_i)\,c$, and from there only through the front speed $r\,c/(k+c)$. When the antibiotic kills the biomass, less nutrient is consumed, the nutrient rises in the seed (its depletion length grows from 1 to 2.6 mm) and the edge moves faster, by 37 % at the nutrient levels of the seed. The effect on the growth is 2 % or less and could not exceed about 10 %, because the growth law $\dot{\alpha} = k_\alpha \phi_{3D}$ counts all biomass, living or dead, and the seed is already near the bound $k_\alpha T^*$. No effect on the seed stress is resolved beyond its grid uncertainty, because the stress is set by the jump of α across the edge of the seed and the edge moves but does not change its height. The local nutrient decides where the antibiotic acts, and the balance of Eq. (9) says how. A species declines at a rate $\alpha^* b_i/(\eta_i \phi_i^2) - c^* A_{ii}/\eta_i$, slow where the nutrient is high and the species is dense, fast where the nutrient is low or the species is thinly present. The same dose therefore removes a species in one part of the biofilm and not in another, which a model without space cannot show. It also removes the species about eight times faster from a mixed seed than from a seed of one species, which a model without the point model's balance cannot show either.

What a composition-dependent mechanics needs. The single path of Eq. (6) follows from the chosen coupling together with the published growth law, and what it says concerns models rather than these examples. With the published equations the composition cannot change the stress by more than the front speed allows, whatever the species and whatever the antibiotic does to them. A model in which the composition is to change the mechanics therefore needs a second path. The coupling presented here is the base on which such a path can be added without touching the hand-over of c^*, α^* and $\sum_i \bar{\phi}_i$, and the runs give an estimate of what each would do. (i) Growth from the living biomass, derived from the energy term (10) (Section 2.5), makes the antibiotic act on the growth directly: the seed growth falls to 0.51 and 0.34 of the published law and the seed stress to 0.84 and 0.70 (Section 4.6). (ii) A front speed tied to the living biomass, which would also stop dead biofilm from spreading after the antibiotic, changes the stress through the sharpness of the edge. In a test on 16^3 (front speed multiplied by the local living fraction) the seed stress rose by 51 % in the separate example and about fourfold in the mixed one. (iii) A stiffness that depends on the species changes the seed stress by 9 to 89 % in a 16^3 test, depending on which species is stiffer. Each of these is a change of the model, and the value of the coupling as presented is that it separates what the published equations give from what such a change would add.

Antibiotic tolerance. In the point model the antibiotic kills fastest where the nutrient is low (Eq. (9)). Experiments on biofilms report the opposite tendency for many antibiotics: cells in nutrient- or oxygen-limited, slowly growing regions are the most tolerant (Walters et al., 2003; Stewart and Franklin, 2008). This is a property of the energy of the point model, not of the coupling, and a point model whose antibiotic term depends on the growth state would reverse the spatial pattern of Section 4.2 without any change of the coupling.

Comparison with experimental scales. The examples are verified, not validated, and their scales compare with measured ones only loosely. Dental plaque grown in situ on enamel reaches about 15 μm after one day and 50 μm after seven (Al-Ahmad et al., 2007). The 20 μm cube of Klempt et al. (2024) matches a one-day plaque, while the 2 mm cube of the examples, chosen for

the element, is 40 times the seven-day thickness. The compressive modulus of in vitro *S. mutans* biofilms is 0.28 kPa (Rmaile et al., 2013), about 30 times $E_0 = 10$ Pa. With one-way mechanics every stress scales with E_0, so the seed stress becomes about 0.02 Pa and that between rigid bars about 0.2 Pa. This is still four to seven orders of magnitude below the pressures that arrest bacterial or tumour growth (6 to 16 kPa and below; Chu et al., 2018; Helmlinger et al., 1997). This is consistent with neglecting the mechanical feedback here, but it is not a test of the model: the growth of the examples is $\alpha - 1 \approx 10^{-3}$, whereas a plaque that doubles its thickness carries a growth of order one. The measured thickness fixes the time scale. The edge of the biofilm moves at most with $r/2 = 50\ \mu\text{m}/T^*$ (Table 2 of Klempt et al., 2024, $c \to 1$); the plaque grows by 15 μm in the first day and by about 6 μm per day from the first to the seventh, so T^* corresponds to about three to nine days. Over that time $k_\alpha = 10^{-3}/T^*$ gives a local growth of about 10^{-3}, so in this model the plaque thickens through the front term, and α describes a small densification of the biomass already present. A validation would need, for one oral multi-species biofilm, the thickness over days and a measured volume growth of the attached film (fixing T^* and k_α), the modulus of the intact film and a measured deformation or stress on its substrate.

Comparison with a measured biofilm volume. Heine et al. (2025) measured the biofilm volume per confocal image of the four five-species biofilms at days 1 to 21. Their composition and live fraction calibrate the five-species point model in a separate work of the authors (in preparation); the volume is not used there. The commensal biofilms lose volume after the first day and recover partly, the dysbiotic ones grow to a plateau, reached at day 6 under flow and at day 15 in static culture. A flat layer of each calibrated community, a quarter of the box high and fed from the opposite face, is run with T^* spanning days 1 to 21 and no constant fitted (Figure 10). With the front speed of the paper the four layers fill the box by day 10 in the same way, so the volume is set by the box, not by the community; the growth law (10) changes it by less than 10^{-3}, because the local growth k_α is small against the front term. At one tenth of the front speed the volume grows by a factor of 1.8 in 20 days, the size of the measured growth, but steadily and alike in the four communities; tying the front speed to $\sum_i \bar{\phi}_i$ separates them through their living fractions (1.4 to 1.65). No variant shows the decline of the commensal biofilms or a plateau before the box is full. Two terms are missing for this comparison: a front speed of the order of $r/10$ that depends on the living biomass, and a removal of biomass by detachment or lysis. The comparison therefore tests the model rather than confirms it, and it names what the next version needs.

Limitations. (i) The front speed depends on the nutrient only, so dead biofilm keeps spreading after the antibiotic, and with the front speed of Klempt et al. (2024) the biofilm fills the cube within $T^* = 1$. At one tenth of that speed (Section 4.7) the seed stress is 1.6 to 3.3 times higher and the antibiotic tips the balance of the separate seed less far, while its effect on the growth stays within 3 %. (ii) Eq. (3) lets dead biomass grow at the same rate as living biomass. (iii) The point model has no space; the composition moves only through the rule for newly reached points, which decides in the maze whether the biofilm recovers. A species-resolved continuum would replace the rule. (iv) The mechanics is coupled one way, as in Klempt et al. (2024); with $E = 10$ Pa the neglected term acts only for coefficients far beyond the paper values (an effect on the growth from about 10^5 on, 16^3 test). (v) Living and dead biomass, and both species, have the same stiffness. (vi) The examples use synthetic data with known parameters, the stress converges slowly with the minmod limiter when the species start mixed (with superbee to a GCI of 6 %), and the extrapolated stress of the neighboured element carries a GCI of 16.5 %. Calibrated data (for example the five-species data of Heine et al., 2025) and a measured stress are needed before the stresses can be validated, not only verified.

Outlook. The next steps are species-resolved transport, so that the composition moves with the biomass, and a front speed and growth tied to the living biomass, with the consequences for the stress estimated above. Further steps are growth laws with a mechanical feedback, calibrated multi-species data in the element, and a realistic tooth or implant geometry. Each can be added to the coupling as presented here, and the verification procedure, several codes, a manufactured solution and grid convergence indices, carries over.

6 Conclusions

This work couples the multi-species point model of Klempt et al. (2026) to the continuum growth model of Klempt et al. (2024) at every integration point, without changing the equations of either model. The point model receives the local nutrient and antibiotic from the fields, $c^* = c_0^* c$ and $\alpha^* = \alpha_0^* a$, and its living volume fraction sets the uptake $g\,\phi_{3D}(\sum_i \bar{\phi}_i)\,c$ of both.

1. In the coupled equations the composition reaches ϕ_{3D}, α and the stress along one path only, through the nutrient and the front speed $r\,c/(k+c)$, because the growth law $\dot{\alpha} = k_\alpha \phi_{3D}$ counts all biomass. The seed growth is bounded by $k_\alpha T^*$ and lies within 10 % of it, and the front speed changes by at most a factor of two. The examples confirm it. The antibiotic changes the growth by 2 % or less and the converged seed stress by no more than its grid uncertainty (6.57 against 6.42×10^{-4} Pa extrapolated in the separate example, GCI 0.1 and 1.4 %), in the cube and around a molar crown. At one tenth of the front speed of the paper the antibiotic changes the growth by at most 3 %. The consequence is that any model in which the composition is to change the mechanics needs a second path: growth or front speed tied to the living biomass, or a species-dependent stiffness. The coupling presented here is the base for such a term. Growth from the living biomass, derived from a single change of the energy, $-\bar{k}_\alpha \alpha \phi_{3D} \sum_i \bar{\phi}_i$, lowers the seed growth with the antibiotic by 49 and 66 % and the seed stress by 16 and 30 % (128^3). A front speed tied to the living biomass raises the seed stress by 51 % to fourfold, a species-dependent stiffness by 9 to 89 % (16^3 tests).

2. The nutrient adjusts within about $4 \times 10^{-8}\,T^*$, depends on the composition only through $\phi_{3D} \sum_i \bar{\phi}_i$ and, with the Thiele modulus $\Lambda^2 = g\ell^2/d = 1$, is depleted over the half edge of the domain but never exhausted. The time link s is equivalent to the viscosities η_i/s and is fixed by the data.

3. In the point model the antibiotic acts through a balance, $c^* A_{ii} \phi_i^2$ against $\alpha^* b_i$, whose decay rate grows as $1/\phi_i^2$: a species that fills its point is removed slowly, the same species thinly present at once. This explains three results of the examples. The local nutrient decides where the antibiotic acts. A mixed seed loses its living biomass about eight times faster than a seed of one species (0.96 to 0.04 against 0.96 to 0.15). The state given to newly reached material, which neither paper fixes, decides in a maze whether the biofilm recovers after the antibiotic (living volume fraction 0.35 against 0.04). The recovery comes from the volume fraction that new points inherit, so the rule must be stated in any such model.

4. The coupling is implemented in a finite-volume code, in a neighboured-element user element in ANSYS and in a Galerkin user element in Abaqus. It is verified by replay of the point model in all three and by a manufactured solution and grid convergence indices in the finite-volume and Galerkin codes; the neighboured element agrees within its band. The front term $-r \frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla \phi_{3D}$ sets the accuracy of the stress: a constant added diffusion gives a quarter of the converged seed stress, and a second-order treatment is needed. With it the growth converges with a GCI of 0.4 to 4.3 % and the seed stress with 0.1 to 1.4 % when the species start apart.

The next steps are models in which the front speed, the growth or the stiffness depend on the living biomass or on the species, with the consequences estimated here. A further step is a measured thickness, modulus and deformation of one oral biofilm, without which the stresses remain verified but not validated.

Data availability

The finite-volume code, the point-model routine, the run scripts, the result files of the examples and the figure scripts are in the public repository `https://github.com/keisuke58/pde-fem-biofilm`. The examples use the published synthetic data of Fritsch et al. (2025); no calibration data are used in this work. The molar crown of Section 4.9 is derived from the Open-Full-Jaw dataset of Gholamalizadeh et al. (2022), licensed CC BY-NC-SA, and the derived surface is redistributed in the repository under the same licence. The user elements depend on the commercial codes and on the element of Rudolf et al. (2025); the material routines are in the repository.

TODO: Confirm with all authors which parts of the element code may be published, and whether a release (DOI) of the repository is made at submission.

Declarations

Author list, contributions, funding and conflicts of interest: to be agreed.

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A Numerical details

A.1 The neighboured element

Three details of the element were corrected, without changing its equations. The surface points take the value of the nearest Gauss point instead of a distance-weighted mean, the Laplacian is used in a conservative form, and the neighbours of the fit are chosen by distance instead of by element number. Before, ϕ_{3D} leaked through the walls of the cube (the surface points took an inverse-square mean, and the column sums of the Laplacian were not zero), and the stencils of the 16^3 and 24^3 grids were asymmetric inside the domain. With a manufactured solution the corrected element still does not converge from 8^3 to 16^3 (L_2 error of ϕ_{3D} 2.2×10^{-3} to 1.8×10^{-3}, of c 8.2×10^{-3} to 1.2×10^{-2}). The MLS stencils of the Gauss points next to a face include the surface points and are ill-conditioned (error of the Laplacian of the manufactured c: 0.88 on 8^3, 65 on 16^3, against $|\nabla^2 c| = 3.5$). Rebuilding these stencils with the neighbours mirrored across the faces (zero flux built in) makes the operators consistent up to the walls. With the weight width proportional to h the element then converges (ϕ_{3D} 9.5×10^{-4} to 3.4×10^{-4}, order 1.5; c 9.0×10^{-4} to 2.2×10^{-4}, order 2.0). In the coupled example the mirrored operators run on 8^3 but become unstable at a corner of the 16^3 grid after the antibiotic onset; they are not used for the results. The explicit update of the element is limited by $\beta \Delta t / h^2 \leq 0.12$ to 0.16 on 16^3 and 24^3 (Bloch analysis of the stencil). The Lax–Wendroff form with the extremum sensor keeps ϕ_{3D} in $[-0.03, 1.01]$ on 64^3 in Python but is unstable at the corners of the supply face on 24^3 in the element, so the element's stresses are quoted from the upwind MLS gradients. The time-step series of the element and the stability analysis will be given in the supplementary material.

B Verification details

B.1 Manufactured solution and mechanics

Sources for ϕ_{3D} and c were built from a manufactured solution (Roache, 2002) by automatic differentiation and checked symbolically (agreement 10^{-14}). The finite-volume and the Galerkin code converge at second order in L_2 (the error plot is in the supplementary material). The observed orders between 8^3, 16^3 and 32^3 are 1.85 and 1.90 for ϕ_{3D} and 2.03 and 2.01 for c in Python, 1.68 and 1.81 for ϕ_{3D} and 2.00 and 2.05 for c in Abaqus. On 32^3 the errors of ϕ_{3D} are 2.0×10^{-4} and 2.9×10^{-4}. For the SUPG form the order of ϕ_{3D} is about 1.5: the largest error sits in a thin layer next to the wall opposite the supply, where $\nabla c \to \mathbf{0}$ and the front direction changes over a few cells. The analytic tangent of the Abaqus element agrees with central differences to 5×10^{-11}. The last growth field of an Abaqus run, solved by the Python mechanics, gives the seed von Mises stress to 1.1 % (no obstacles) and 0.5 % (obstacles with frictionless contact).

B.2 The front term and the agreement of the codes

Four treatments of the front term were compared, none of which changes Eqs. (1) to (3). They are (1) a constant added diffusion $rh/4$ (first order; the element as delivered needs it to stay stable) and (2) TVD upwinding (MUSCL with the minmod limiter; Python). The other two are (3) SUPG with $\tau = \frac{h}{2|v|}(\coth Pe - 1/Pe)$ on the strong residual (Abaqus) and (4) for the explicit element, either the Lax–Wendroff term $\frac{\Delta t}{2}\, v \cdot \nabla(v \cdot \nabla \phi_{3D})$ with an added diffusion switched on by an extremum sensor, or upwind-biased MLS gradients. Figure 11 shows the seed means of $\alpha - 1$ and of the von Mises stress at $T^* = 1$ of the separate example with antibiotic against the grid for every scheme and code. The growth variable converges for every scheme, and the codes agree within their grid-convergence bands: Python extrapolates to 9.0 to 9.8×10^{-4}, Abaqus to 8.8 and 9.45×10^{-4}, the element to 8.7×10^{-4} with upwind gradients. The stress does not converge with the constant added diffusion: on 16^3 it is about a quarter of the converged value (1.3 to 1.7 against 5.9 to 6.3×10^{-4} Pa from the two second-order schemes of two codes). On 96^3 it is still rising. Treatments 2 to 4 give the same stress within a few per cent from 32^3 on (Python TVD 5.90 and 5.98×10^{-4} Pa on 32^3 and 64^3, Abaqus SUPG 6.27 and 6.22×10^{-4} Pa on 32^3 and 48^3, GCI 1.2 %). The element with upwind gradients gives 2.14, 4.56 and 5.15×10^{-4} Pa on 8^3, 16^3 and 24^3, extrapolated 5.8×10^{-4} Pa (GCI 16.5 %), within the band of the second-order schemes. Its MLS operators over 30 neighbours give it a resolution of about one element, not one Gauss-point spacing (with the same added diffusion, the element on 16^3 and Python on 32^3 differ by 1 % in $\alpha - 1$ and 5 % in the stress).

B.3 Time step, regularisation and stability

In the element codes the point model advances in every increment (0.00125 in T^*; Abaqus: observed order 1.1 in time, error against the extrapolated value 0.8 % in $\alpha - 1$ and 2.2 % in the stress). In the Python code the point model and $\sum_i \bar{\phi}_i$ are updated once per outer step ΔT^*, inside which ϕ_{3D}, α and the fields take explicit RK4 steps. The composition and $\alpha - 1$ change by less than 3 and 0.3 % between outer steps of 0.025 and 0.0025; the stress does not. On 64^3 the separate example with antibiotic gives 5.68, 5.82, 5.94 and 6.00×10^{-4} Pa at $\Delta T^* = 0.025$, 0.01, 0.005 and 0.0025, the mixed one 3.23, 3.09, 2.90 and 2.82×10^{-4} Pa. The sensitivity comes from the front direction: after the antibiotic onset $\sum_i \bar{\phi}_i$ is small, ∇c is weak and $\mathbf{n}_c$ responds to small changes of the consumption (a regularisation $\varepsilon^2 = 10^{-4}$ instead of 10^{-6} changes the mixed stress by 30 %, 10^{-8} by 0.1 %). The inner step, a trapezoidal update of $\sum_i \bar{\phi}_i$, consumption by newly reached points at once and the one-increment lag of $\sum_i \bar{\phi}_i$ each change the stress by less than 1 %. The grid series are therefore run at $\Delta T^* = 0.0025$. The explicit inner step is limited by $\beta \Delta t / h^2 \leq 0.15$ and the Courant number of the front $\mathrm{Co} \leq 0.4$. On 192^3 both limits are close

765 at the same time and the step diverged, which the combined condition $12D + 2\,\mathrm{Co} \leq 2.2$ with
766 $D = \beta\Delta t/h^2$ removes without changing the step on 32^3 to 128^3.

767 B.4 Grid convergence table

Table 5: Grid convergence (Python, TVD with the minmod limiter unless another limiter is stated, outer step 0.0025): seed means at $T^* = 1$, observed order p, extrapolated value (extrap.) and GCI (Celik et al., 2008), from 64^3, 128^3, 192^3; –: not run; antib.: with antibiotic. Mixed: equal amounts unless the start fractions (ϕ_1, ϕ_2) are given.

example	quantity	32^3	64^3	128^3	192^3	p	extrap.	GCI
separate, antib.	$\alpha - 1$ [10^{-4}]	8.453	8.931	9.112	9.152	1.7	9.19	0.5 %
separate, antib.	von Mises [10^{-4} Pa]	5.523	6.000	6.284	6.349	1.6	6.42	1.4 %
separate	$\alpha - 1$ [10^{-4}]	8.729	9.014	9.110	9.135	1.5	9.17	0.4 %
separate	von Mises [10^{-4} Pa]	7.440	6.730	6.585	6.571	3.1	6.57	0.1 %
mixed, antib.	$\alpha - 1$ [10^{-4}]	7.42	8.04	8.32	8.42	0.77	8.71	4.3 %
mixed, antib.	von Mises [10^{-4} Pa]	2.00	2.82	3.30	3.52	0.43	4.68	41 %
mixed	$\alpha - 1$ [10^{-4}]	7.46	8.08	8.35	8.45	0.77	8.74	4.2 %
mixed	von Mises [10^{-4} Pa]	2.13	3.08	3.65	3.89	0.55	4.86	31 %
mixed $(0.25, 0.30)$, antib.	$\alpha - 1$ [10^{-4}]	7.42	8.04	8.31	8.40	0.93	8.61	3.1 %
mixed $(0.25, 0.30)$, antib.	von Mises [10^{-4} Pa]	2.01	2.96	4.01	4.35	1.0	5.01	19 %
mixed, antib., superbee	$\alpha - 1$ [10^{-4}]	7.79	8.37	8.49	8.51	2.5	8.52	0.2 %
mixed, antib., superbee	von Mises [10^{-4} Pa]	2.39	3.28	3.59	3.67	1.2	3.81	4.6 %
mixed, superbee	$\alpha - 1$ [10^{-4}]	7.84	8.41	8.53	8.55	2.7	8.55	0.1 %
mixed, superbee	von Mises [10^{-4} Pa]	2.57	3.64	3.99	4.07	1.7	4.16	2.5 %
separate, antib., superbee	$\alpha - 1$ [10^{-4}]	–	9.161	9.209	–			
separate, antib., superbee	von Mises [10^{-4} Pa]	–	6.122	6.338	–			
mixed $(0.25, 0.30)$, antib., superbee	von Mises [10^{-4} Pa]	2.39	3.46	4.41	4.52	2.8	4.57	1.5 %
mixed, antib., MC	$\alpha - 1$ [10^{-4}]	7.69	8.23	8.40	8.47	0.69	8.69	3.2 %
mixed, antib., MC	von Mises [10^{-4} Pa]	2.26	3.07	3.44	3.61	0.43	4.5	31 %
mixed, antib., WENO5	von Mises [10^{-4} Pa]	2.48	3.25	3.51	3.63	0.49	4.2	18 %
mixed, antib., van Leer	$\alpha - 1$ [10^{-4}]	7.59	8.17	8.38	8.46	0.92	8.62	2.5 %
mixed, antib., van Leer	von Mises [10^{-4} Pa]	2.16	2.98	3.40	3.58	0.60	4.2	22 %

768 The stress of the mixed examples converges slowly because the edge that the front term
769 moves is thin. The front speed is $v = r\,c/(k + c) \approx 4.9$ mm$/T^*$ at the nutrient levels of the seed,
770 so the edge is about $\beta/v \approx 0.004$ mm wide, and the cell Péclet number $vh/(2\beta)$ is 1.9 on 128^3
771 and 1.3 on 192^3; it falls below one only on 256^3. Without the front term the mixed seed stress
772 changes by 0.3 % between 32^3 and 64^3 (6.07 and 6.06×10^{-4} Pa), and $\varepsilon^2 = 10^{-8}$ gives the same
773 values as 10^{-6}, so neither the remaining terms nor the regularisation cause the slow convergence.
774 Resolving the edge of ϕ_{3D} alone is not enough. In a two-level test on 32^3 cells, with the cells near
775 the edge refined by a factor of two, the mixed seed stress with antibiotic is 2.7×10^{-4} Pa, against
776 2.24 on uniform 32^3 and 3.23×10^{-4} Pa on uniform 64^3; it reaches the 64^3 value (3.22×10^{-4} Pa)
777 only when the nutrient, and with it the front direction $\mathbf{n}_c$, is solved on the finer grid as well. The
778 uptake $g\,\phi_{3D}(\sum_i \bar{\phi}_i)\,c$ is concentrated at the edge, so the nutrient gradient that steers the front
779 has to be resolved there too. On 256^3, where the cell Péclet number is 0.96, the seed stress of
780 the mixed examples with antibiotic is 3.63 (equal shares) and 4.50×10^{-4} Pa (shares of Fritsch
781 et al.); these two runs keep $\dot{c}$, the coarser ones use its limit, which differs by at most 0.1 %. The
782 stress still rises with the grid, but by half as much from 192^3 to 256^3 ($+0.11 \times 10^{-4}$ Pa) as from
783 128^3 to 192^3. On the three finest grids (128^3, 192^3, 256^3) the observed order rises to 0.93 and
784 1.3, the extrapolated stress is 4.0 and 4.8×10^{-4} Pa and the GCI 13 and 9 %, against 31–41 %
785 on 64^3–192^3. The growth $\alpha - 1$ is 8.47 and 8.45×10^{-4}, with a GCI of 3 %. The mixed stress
786 with antibiotic is therefore known to about 10 %. Without antibiotic (equal shares) the seed

stress on 256^3 is 4.02×10^{-4} Pa, with $\dot{c}$; on 128^3, 192^3 and 256^3 the observed order is 0.85, the extrapolated stress 4.48×10^{-4} Pa and the GCI 14 %, and the growth is 8.50×10^{-4}. On 256^3 the stress without antibiotic is 11 % above that with it, and the extrapolated values differ by 12 % (4.48 against 4.0), less than the two bands together (14 and 13 %). As on the coarser grids, no effect of the antibiotic on the mixed stress is resolved, while the growth differs by 0.4 %.

The limiter is the other part. The superbee limiter keeps the edge sharper than minmod at the same grid. With it the mixed seed stress with antibiotic is 2.39, 3.28, 3.59 and 3.67×10^{-4} Pa on 32^3 to 192^3 (last rows of Table 5): observed order 1.2, extrapolated 3.81×10^{-4} Pa, GCI 4.6 %, against 0.43 and 41 % with minmod on the same grids. Superbee on 192^3 is close to minmod on 256^3 (3.67 against 3.63×10^{-4} Pa), and its extrapolation lies within the band of the minmod extrapolation from 128^3 to 256^3 (4.0×10^{-4} Pa, GCI 13 %). Superbee on 256^3 gives 3.76×10^{-4} Pa, below the extrapolated 3.81; from 128^3, 192^3, 256^3 it extrapolates to 3.95×10^{-4} Pa (order 1.3, GCI 6.3 %), so the superbee series is near but not fully in its asymptotic range, and the band on the finest triple is the one to quote. The minmod extrapolation from 64^3 to 192^3 (4.68×10^{-4} Pa) is therefore too high: on those grids minmod is not yet in its asymptotic range. The growth changes by 1 % (8.51 against 8.42×10^{-4} on 192^3). A compressive limiter can also steepen an edge beyond the solution, so the two limiters together, not superbee alone, bound the mixed stress. In the separate example with antibiotic, where minmod converges, superbee gives 6.12 and 6.34×10^{-4} Pa on 64^3 and 128^3, within 1.3 % of the minmod extrapolation (6.42), and the growth within 0.2 % of it: where the edge is resolved, the limiter does not change the answer. In the mixed example without antibiotic superbee gives 2.57, 3.64, 3.99 and 4.07×10^{-4} Pa on 32^3 to 192^3 (observed order 1.7, extrapolated 4.16, GCI 2.5 %), and 4.18×10^{-4} Pa at the resolution of 256^3 (two-level grid on a 128^3 base, which matches uniform 256^3 with antibiotic to 0.1 %; extrapolated from 128^3 to 256^3: 4.26, GCI 2.4 %); with minmod the difference to the run with antibiotic disappears within the two bands of 31 and 41 %. With the start fractions $(0.25, 0.30)$ superbee gives 2.39, 3.46, 4.41 and 4.52×10^{-4} Pa and 4.62 at the resolution of 256^3 (from 128^3 to 256^3: order 1.2, extrapolated 4.87, GCI 6.7 %; the 1.5 % of 64^3 to 192^3 underestimated the band), against an extrapolated 5.01 with a band of 19 % for minmod. The MC and van Leer limiters, which lie between minmod and superbee in how much they diffuse, give mixed stresses close to minmod on every grid (MC: observed order 0.43, GCI 31 %, extrapolated 4.5×10^{-4} Pa; van Leer: 0.60, 22 %, 4.2×10^{-4} Pa); the growth of all four limiters agrees within 1 % on 192^3; only superbee reaches the asymptotic range on 64^3 to 192^3. Figure 12 collects the series. The WENO5 reconstruction (Jiang and Shu, 1996) gives 2.48, 3.25, 3.51 and 3.63×10^{-4} Pa on 32^3 to 192^3: above superbee on the coarsest grid, between MC and superbee on the finer ones, and it converges as slowly as minmod (order 0.49, GCI 18 %). Its high order holds only where the solution is smooth; at the nearly discontinuous edge it falls back to first order like the TVD limiters.

Supplementary material

The supplementary material will provide the L_2 error plot of the manufactured solution, the discretisation table of the three codes, and the time-step series of the neighboured element and of the Galerkin code. It will also provide the stability analysis of the explicit element (Bloch analysis of the stencil, step limits $\beta \Delta t / h^2 \leq 0.12$ to 0.16), the sensitivities to $\phi_{\min}$, d, ε, f and the start rule, and the time series of the three antibiotic schedules (Section 4.3). The test of the mirrored operators of the element and the animation of the maze will be provided as well.

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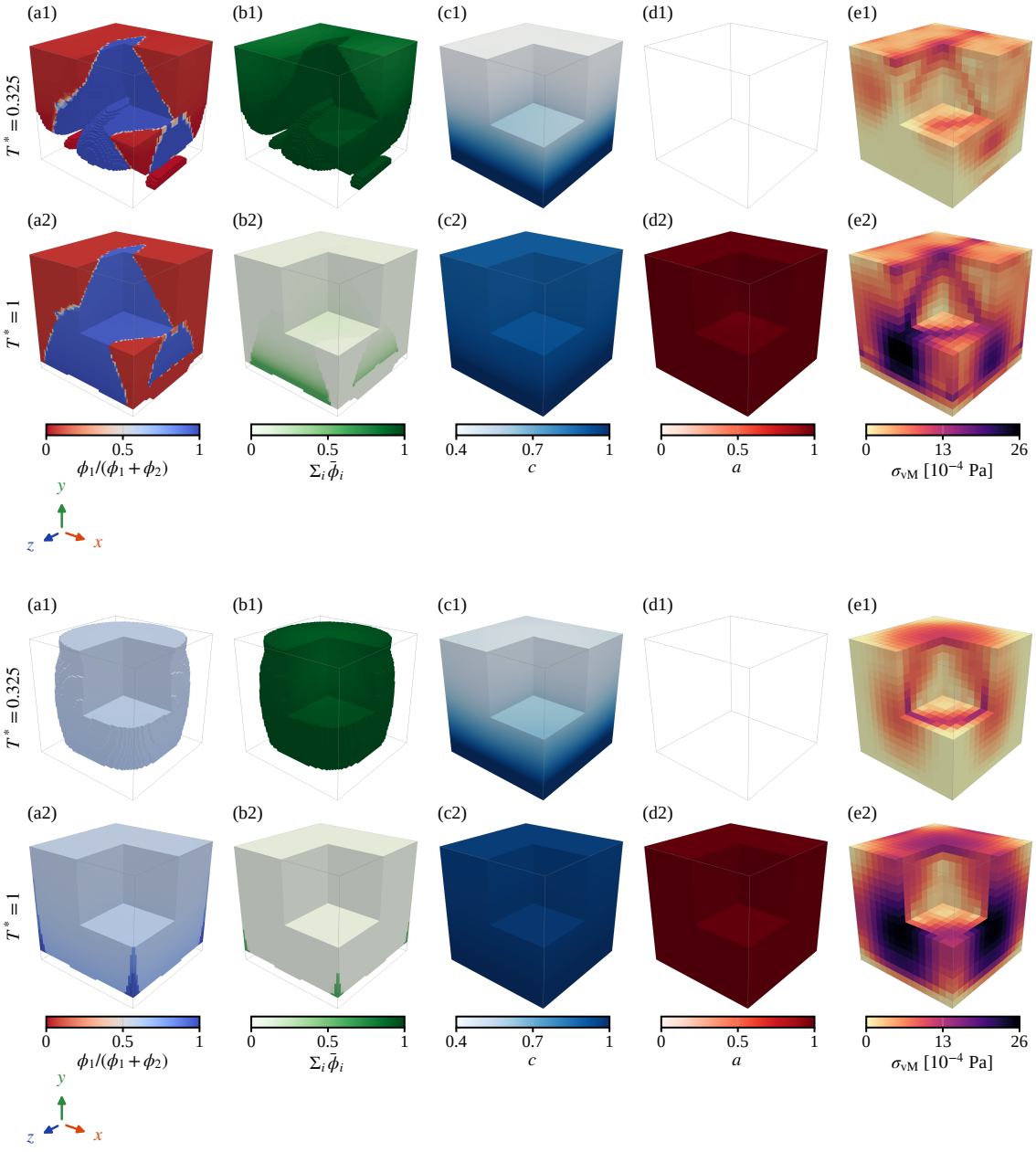


Figure 3: Separate (upper half) and mixed (lower half) example with antibiotic, Python, 64^3, one octant removed, at $T^* = 0.325$ (first row of each half) and $T^* = 1$ (second row): (a) share $\phi_1/(\phi_1+\phi_2)$, (b) living volume fraction $\sum_i \bar{\phi}_i$, (c) nutrient c, (d) antibiotic a, (e) von Mises stress. In the mixed example the living volume fraction falls almost everywhere after the antibiotic.

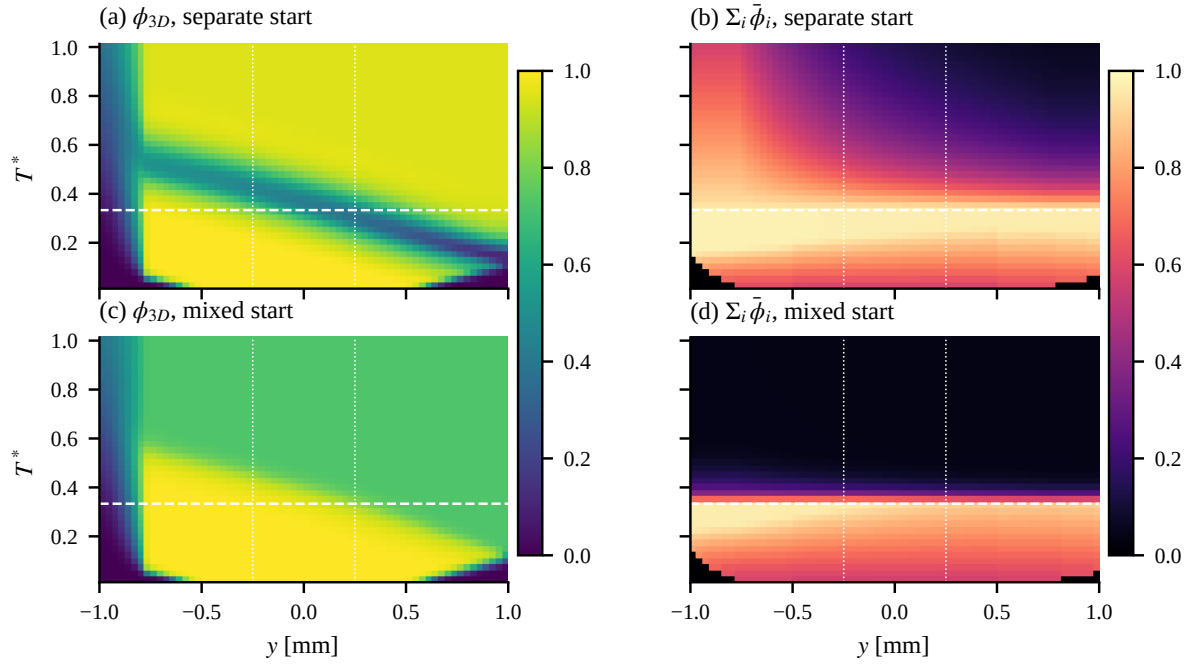


Figure 4: Space-time diagram along the line $x = z = 0$ through the seed (64^3, outer step 0.025, with antibiotic): (a, c) ϕ_{3D} and (b, d) living volume fraction $\sum_i \bar{\phi}_i$ over y (nutrient supplied at $y = -1$ mm) and T^*, separate (upper row) and mixed start (lower row). Dashed: onset of the antibiotic, $T^* = 1/3$; dotted: the seed.

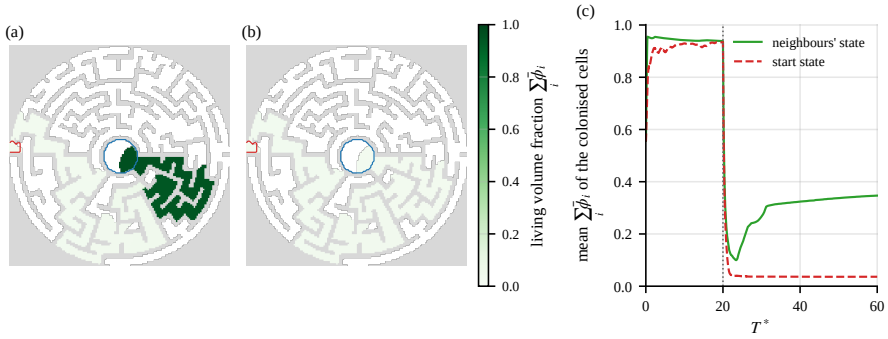


Figure 5: The maze with the antibiotic held at the entrance from $T^* = 20$, at $T^* = 60$: living volume fraction $\sum_i \bar{\phi}_i$ where $\phi_{3D} \geq 0.05$ when newly reached cells (a) take the state of their colonised neighbours and (b) take the start state of the colony; walls grey, nutrient source outlined in blue, antibiotic source in red. (c) Living volume fraction averaged over the colonised cells against T^* for both rules; the antibiotic starts at the dotted line.

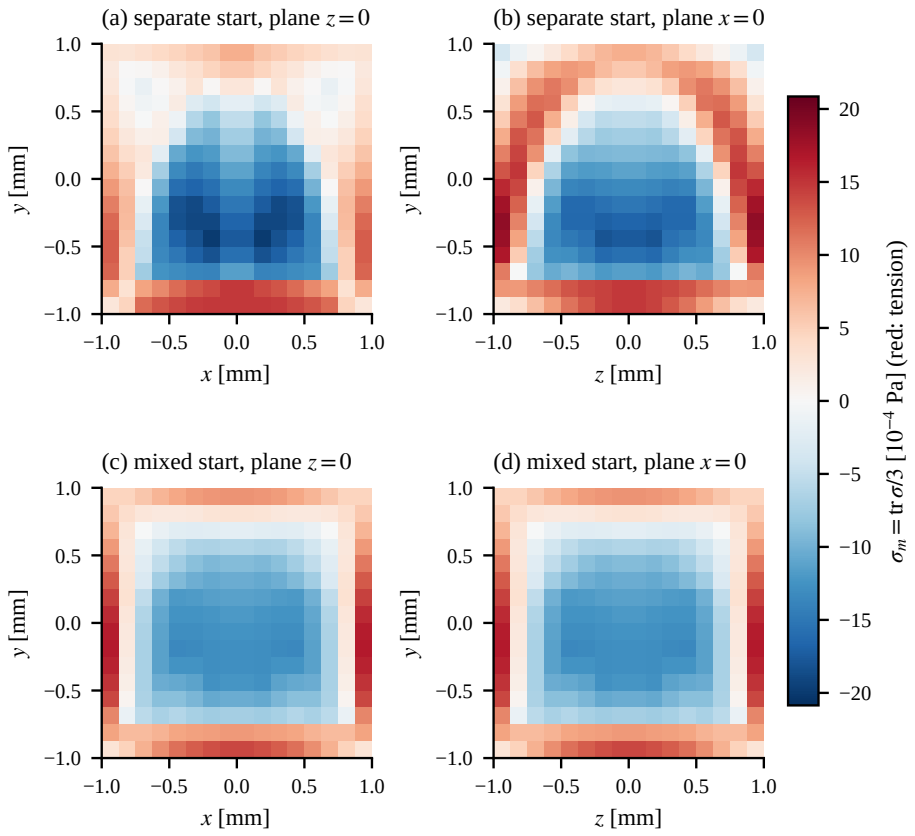


Figure 6: Hydrostatic stress $\sigma_m = \mathrm{tr}\,\boldsymbol{\sigma}/3$ at $T^* = 1$ (red: tension, blue: compression) on the mid-planes $z = 0$ and $x = 0$, separate (a, b) and mixed (c, d) example with antibiotic; fields on 64^3 (outer step 0.025), stress on the 16^3 mesh.

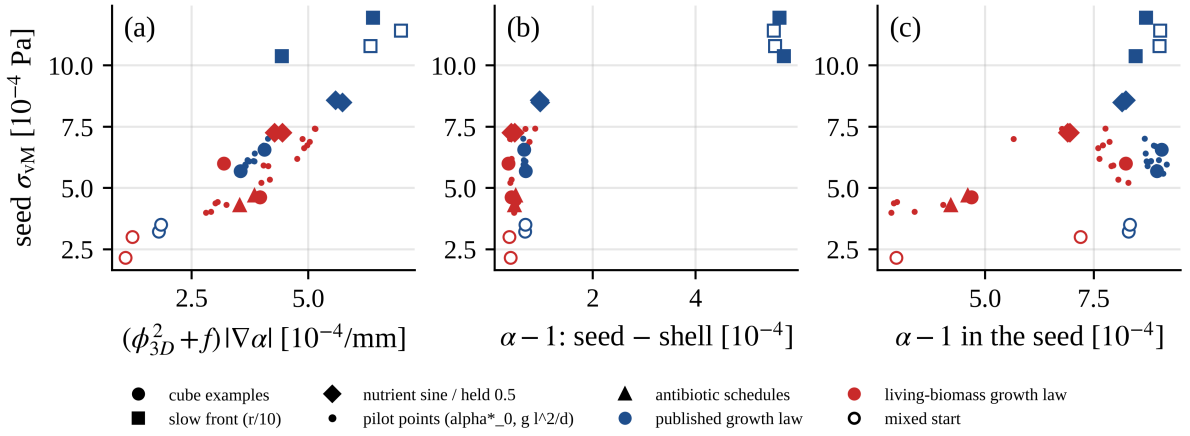


Figure 7: Seed von Mises stress at $T^* = 1$ of 51 runs on 64^3 against (a) the stiffness-weighted gradient of the growth, $(\phi_{3D}^2 + f)\,|\nabla\alpha|$, averaged over the seed and the shell of one element (0.25 mm) around it, (b) the difference of $\alpha - 1$ between the seed and that shell, and (c) the growth $\alpha - 1$ of the seed. Blue: published growth law; red: growth of the living biomass (Section 2.5); open symbols: mixed start. Correlation with the stress in (a): 0.94 and 0.90.

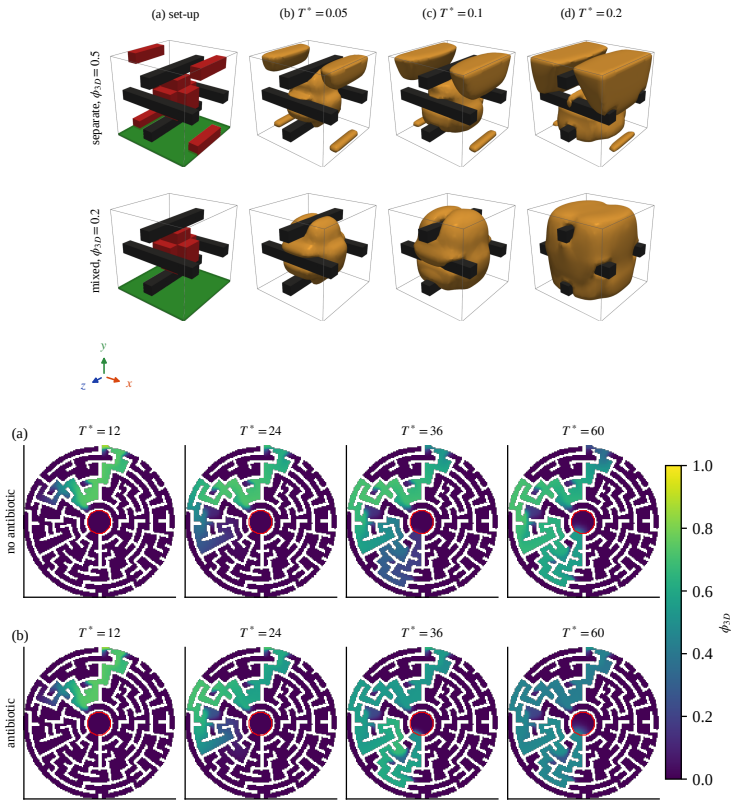


Figure 8: Upper half: growth against four rigid bars (dark grey) after Fig. 9 of Klempt et al. (2024), Python, 32^3, no antibiotic. (a) Set-up: start biomass (red) and nutrient supply on the face $y = -1$ mm (green). (b)–(d) Surface $\phi_{3D} = 0.5$ (separate example) and 0.2 (mixed example, whose front carries lower values of ϕ_{3D}) at $T^* = 0.05$, 0.1 and 0.2. Lower half: the maze of Fig. 11 of the same paper with the coupled model (two dimensions, 4 μm cells): ϕ_{3D} at $T^* = 12$, 24, 36 and 60, (a) without and (b) with the antibiotic at the entrance from $T^* = 20$. Walls white, nutrient source outlined in red (in Figure 5 it is outlined in blue).

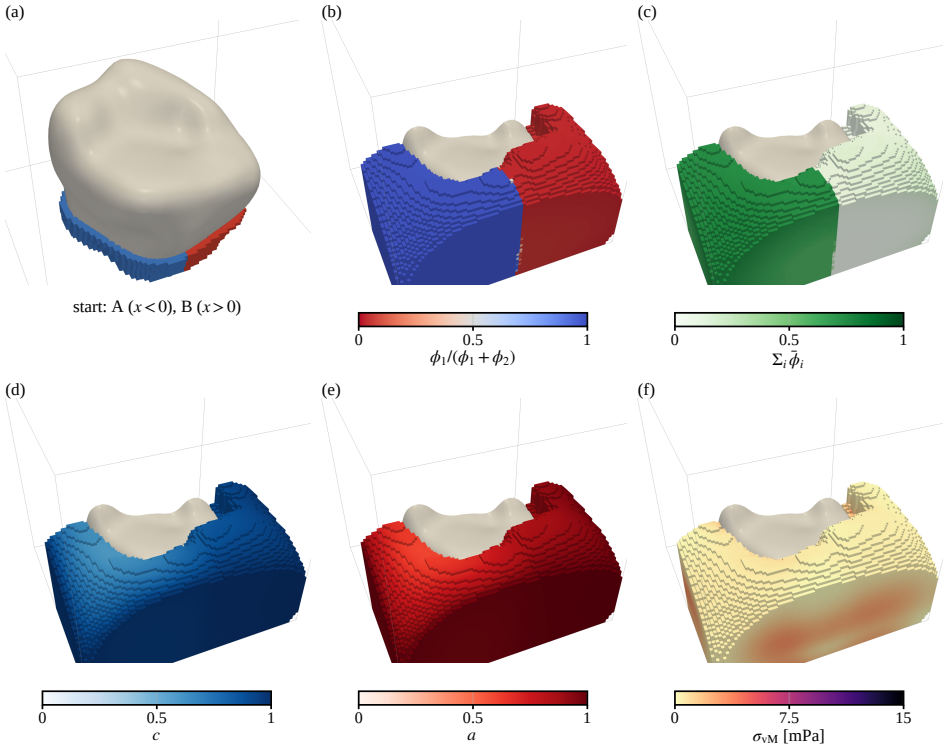


Figure 9: The coupled model around the crown of a human molar (Gholamalizadeh et al., 2022), 80^3 cells, with the antibiotic from $T^* = 1/3$. (a) Start: the cervical ring, species A on the half $x < 0$ (blue), B on $x > 0$ (red). (b)–(f) The film $\phi_{3D} > 0.5$ at $T^* = 1$, coloured by (b) the share $\phi_1/(\phi_1 + \phi_2)$, (c) the living volume fraction $\sum_i \bar{\phi}_i$, (d) the nutrient c, (e) the antibiotic a and (f) the von Mises stress.

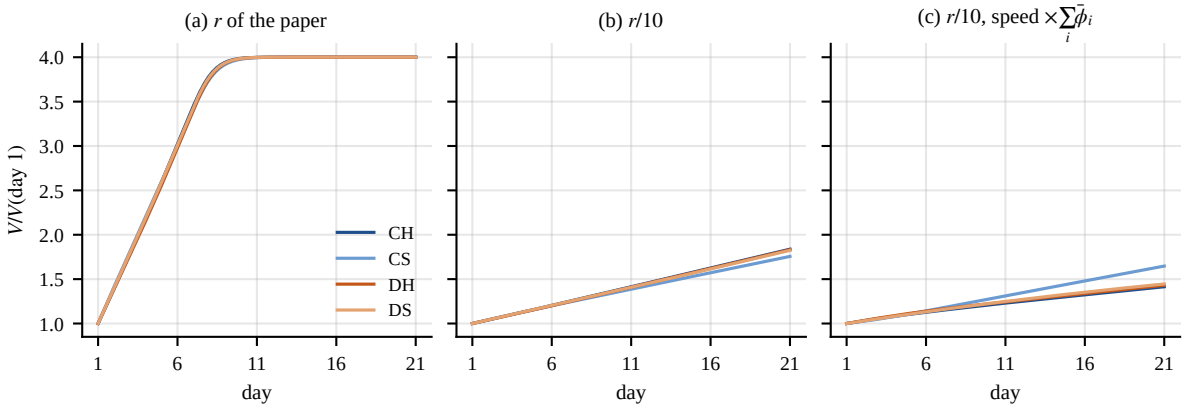


Figure 10: Biofilm volume of a flat layer with the calibrated five-species communities (commensal and dysbiotic, under flow (H) and static (S)), relative to day 1: (a) front speed of the paper, (b) one tenth of it, (c) one tenth multiplied by the local living fraction $\sum_i \bar{\phi}_i$. 32^3 cells, outer step 0.0025. The measured volumes of Heine et al. (2025) fall after day 1 for the commensal and rise to a plateau of about 1.3 to 1.6 for the dysbiotic biofilms (their Fig. 2A).

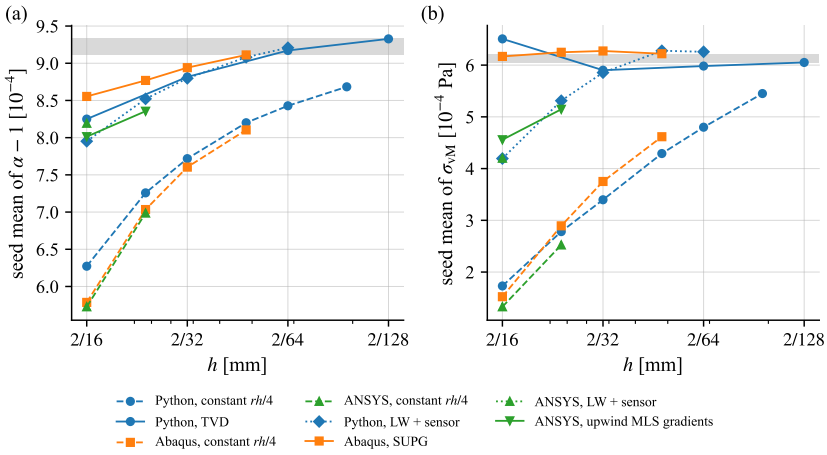


Figure 11: Separate example with antibiotic: seed means of (a) $\alpha - 1$ and (b) the von Mises stress at $T^* = 1$ against the grid, one line per scheme and code (dashed: constant added diffusion $rh/4$; solid and dotted: the second-order and the stabilised explicit forms). Grey band: the finest grids of the two second-order schemes. These runs start newly reached points from the seed composition (the alternative rule of Section 4.4) at a field step of 0.00125.

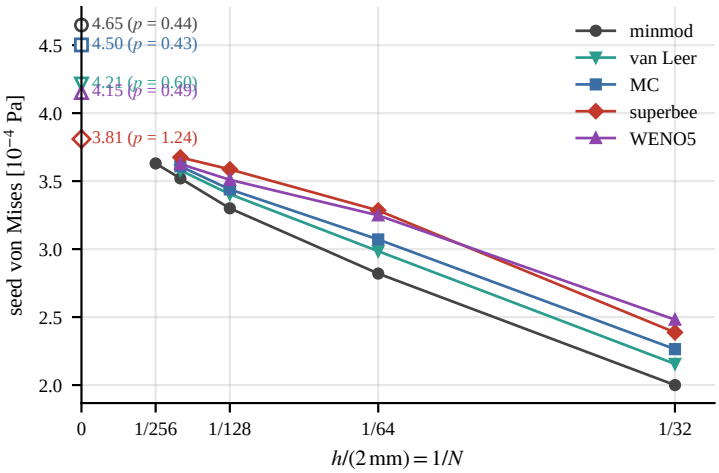


Figure 12: Seed von Mises stress of the mixed example with antibiotic against the grid (N^3 cells, outer step 0.0025) for five reconstructions of the front term. Open symbols: extrapolation from 64^3, 128^3, 192^3 with the observed order p (Celik et al., 2008).